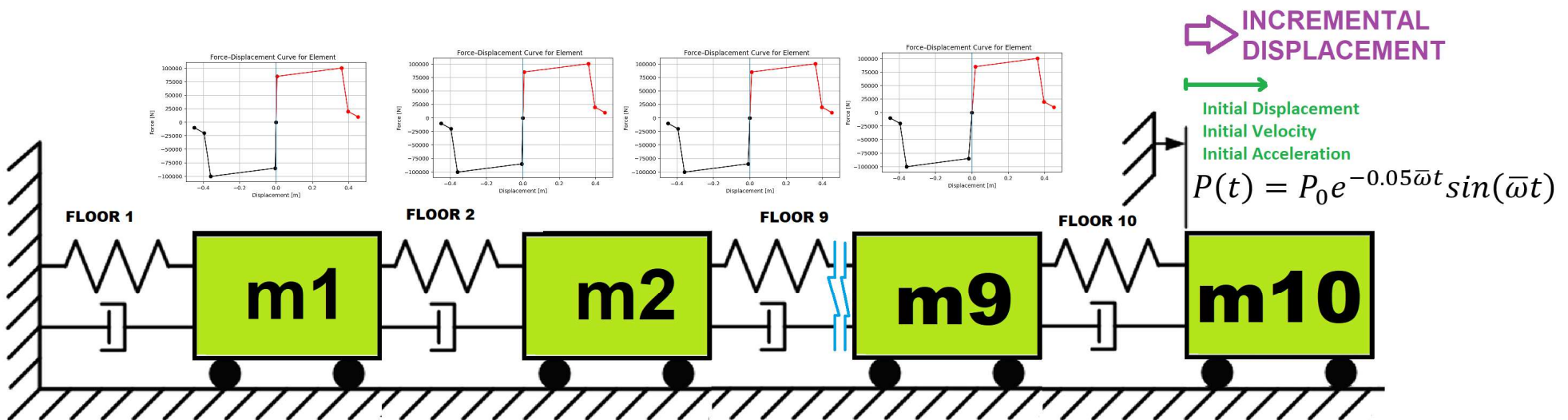


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

**FRAGILITY ANALYSIS BASED ON
ACCELERATION, STRUCTURAL DUCTILITY
DAMAGE INDEX, ENERGY DISSIPATION
CAPACITY INDEX, EQUIVALENT VISCOUS
DAMPING RATIO WITH NONLINEAR DYNAMIC
ANALYSIS OF A MULTI-DEGREE-OF-FREEDOM
(MDOF) SYSTEM UTILIZING 200 GROUND
MOTIONS IN OPENSEES**

WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

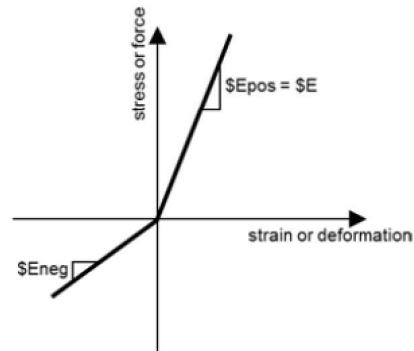
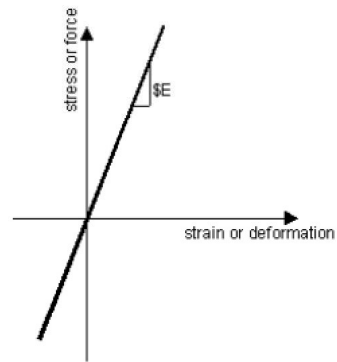
Δ_d = Lateral Displacement from Dynamic Analysis

Δ_y = Lateral Yield Displacement from Pushover Analysis

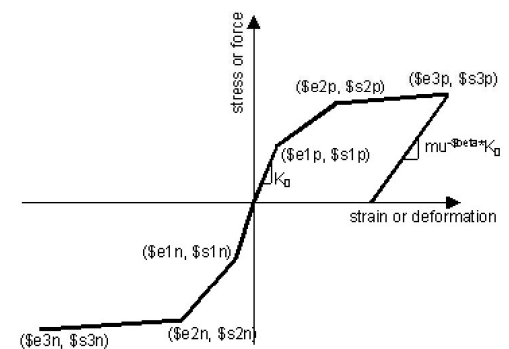
Δ_u = Lateral Ultimate Displacement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

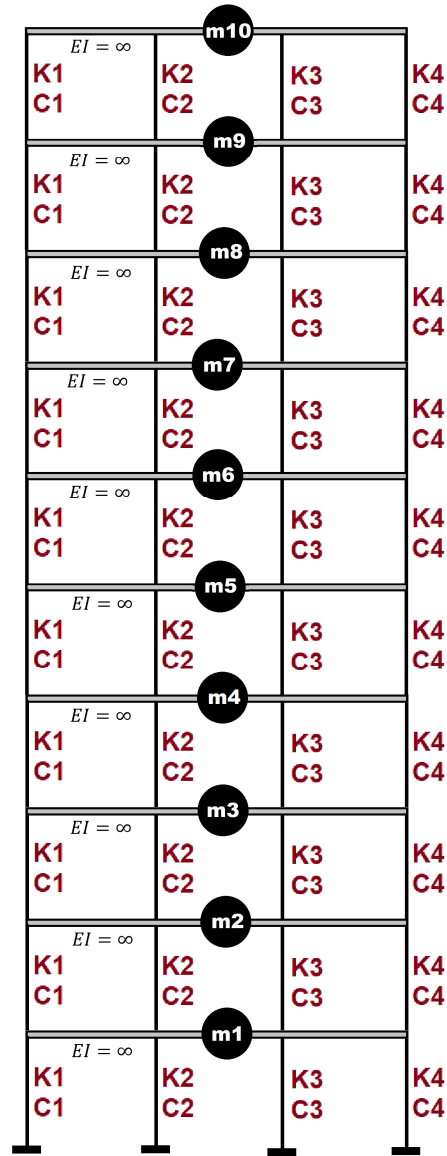
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL



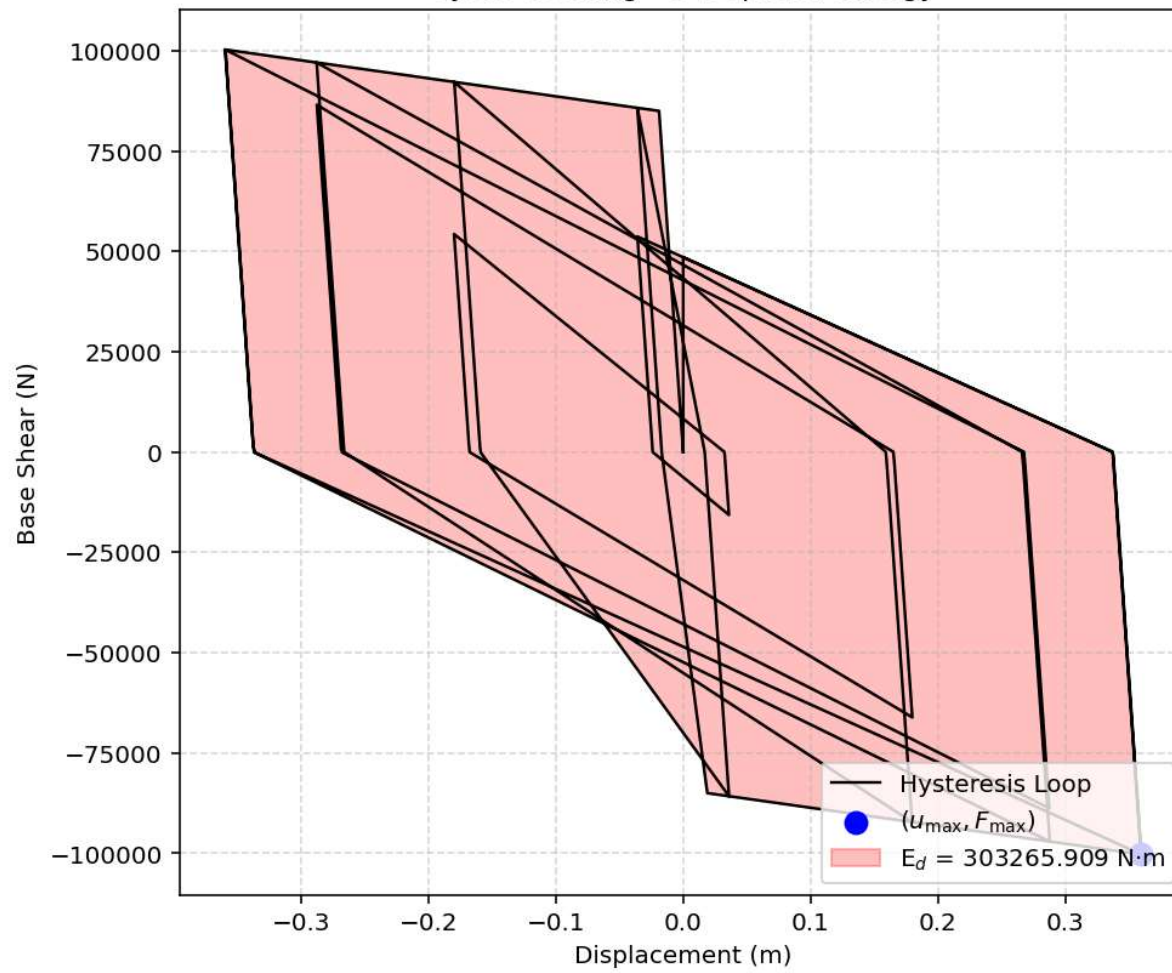
$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

Δ_d = Lateral Displacement from Dynamic Analysis

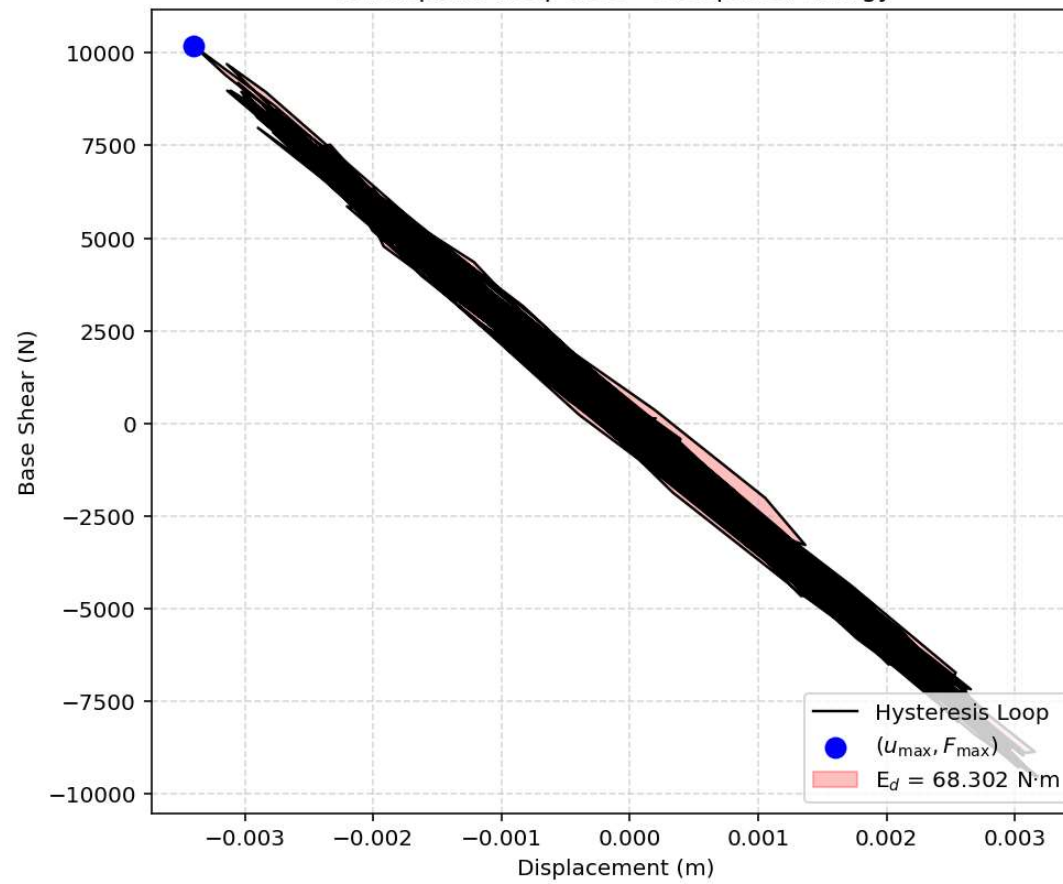
Δ_y = Lateral Yield Displacement from Pushover Analysis

Δ_u = Lateral Ultimate Displacement from Pushover Analysis

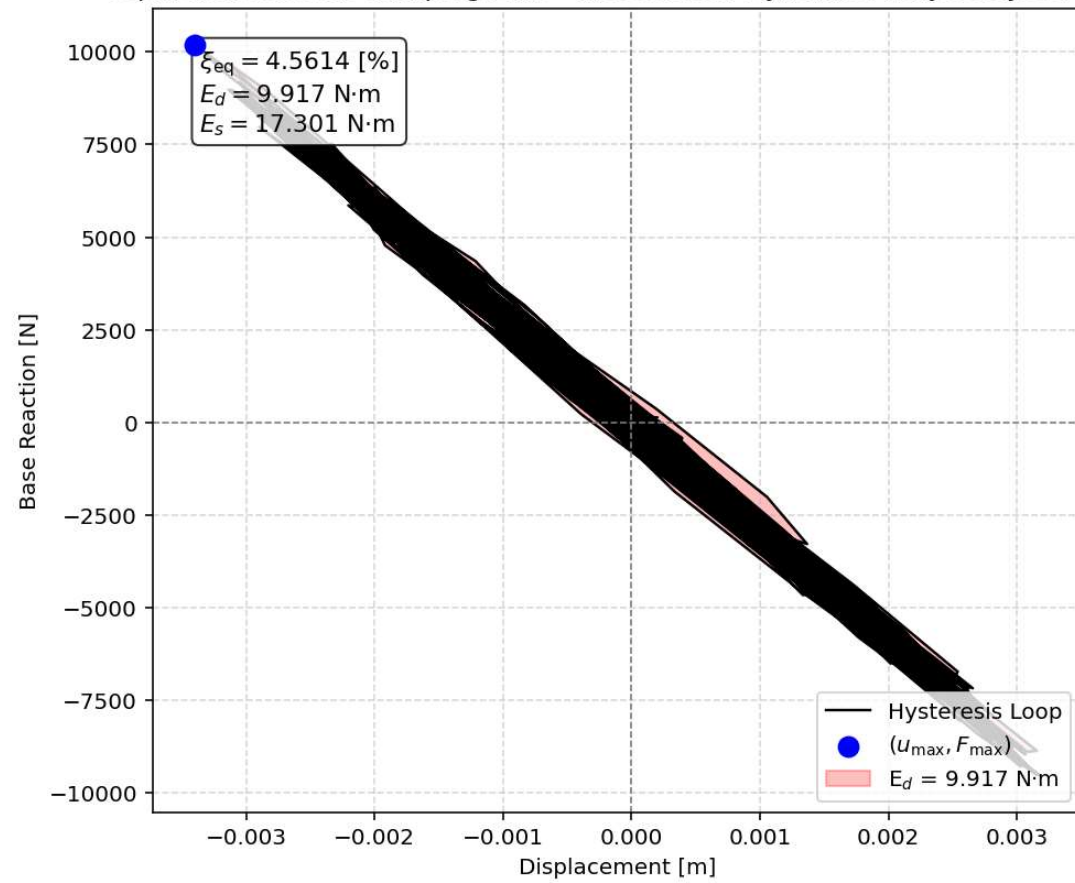
Cyclic Loading - Dissipated Energy



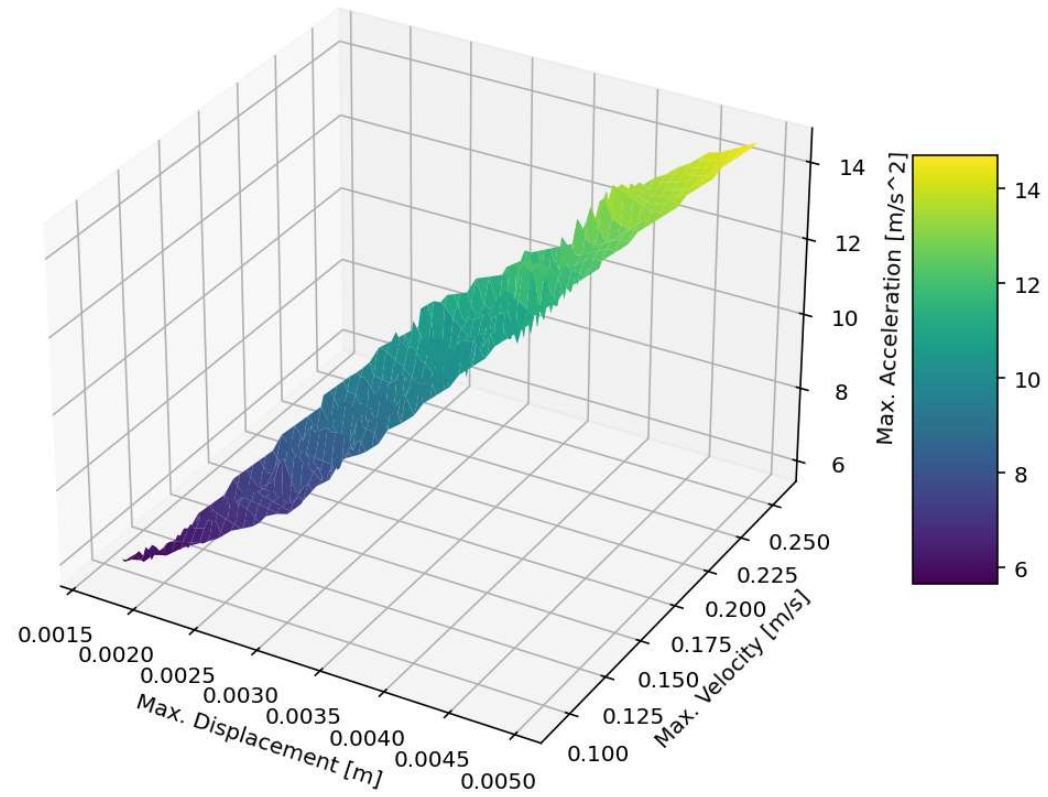
Earthquake Response - Dissipated Energy

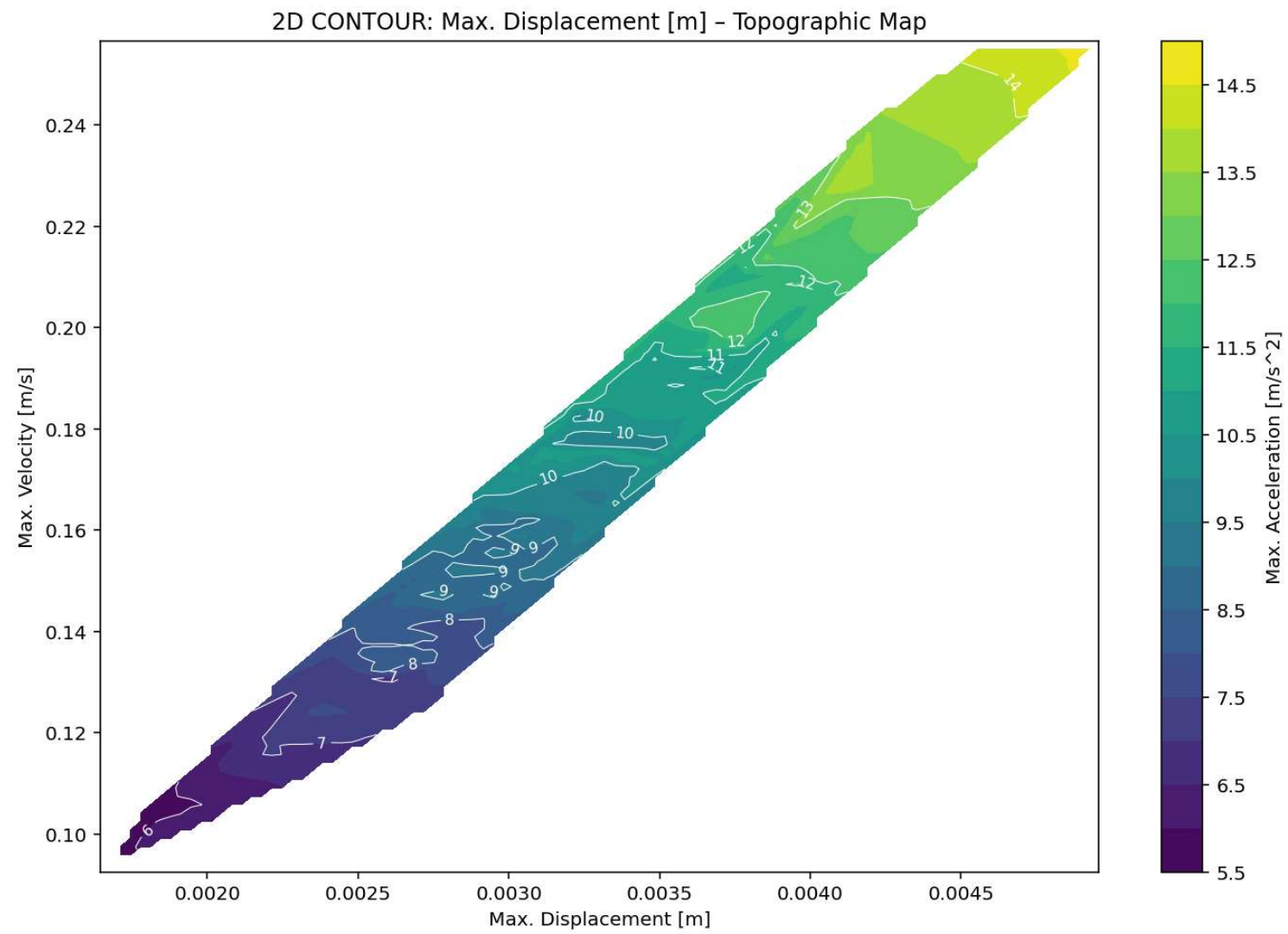


Equivalent viscous damping ratio - Incremental Dynamic Analysis Hysteresis

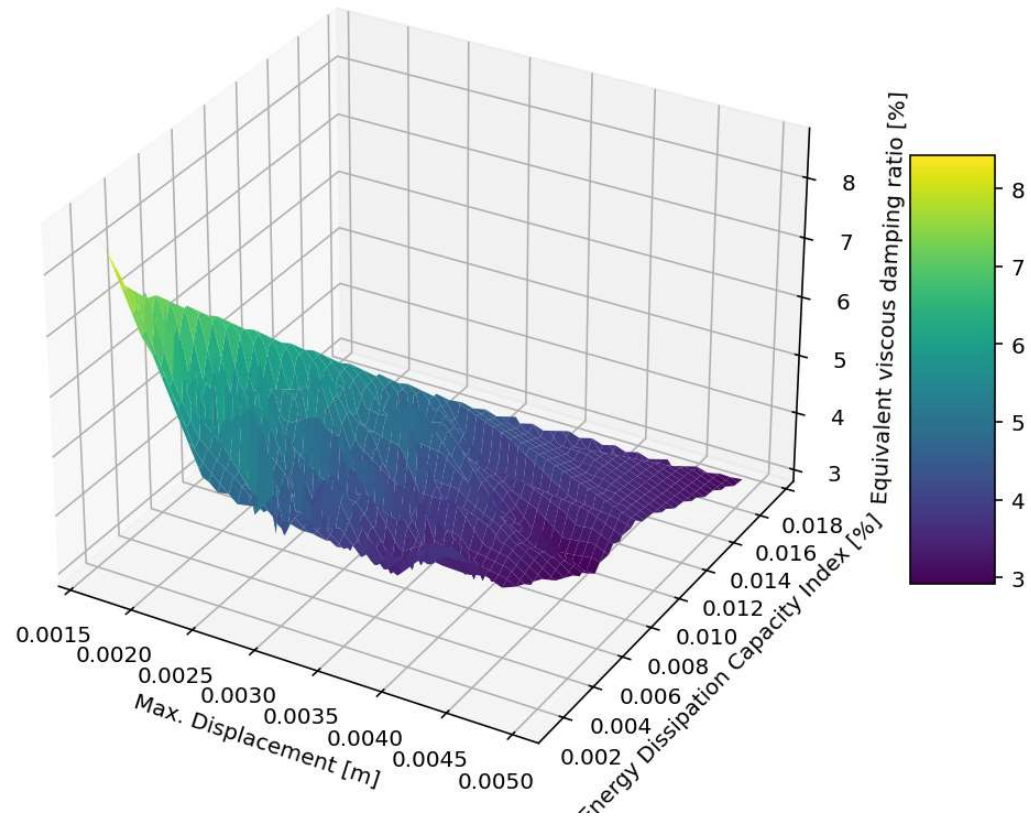


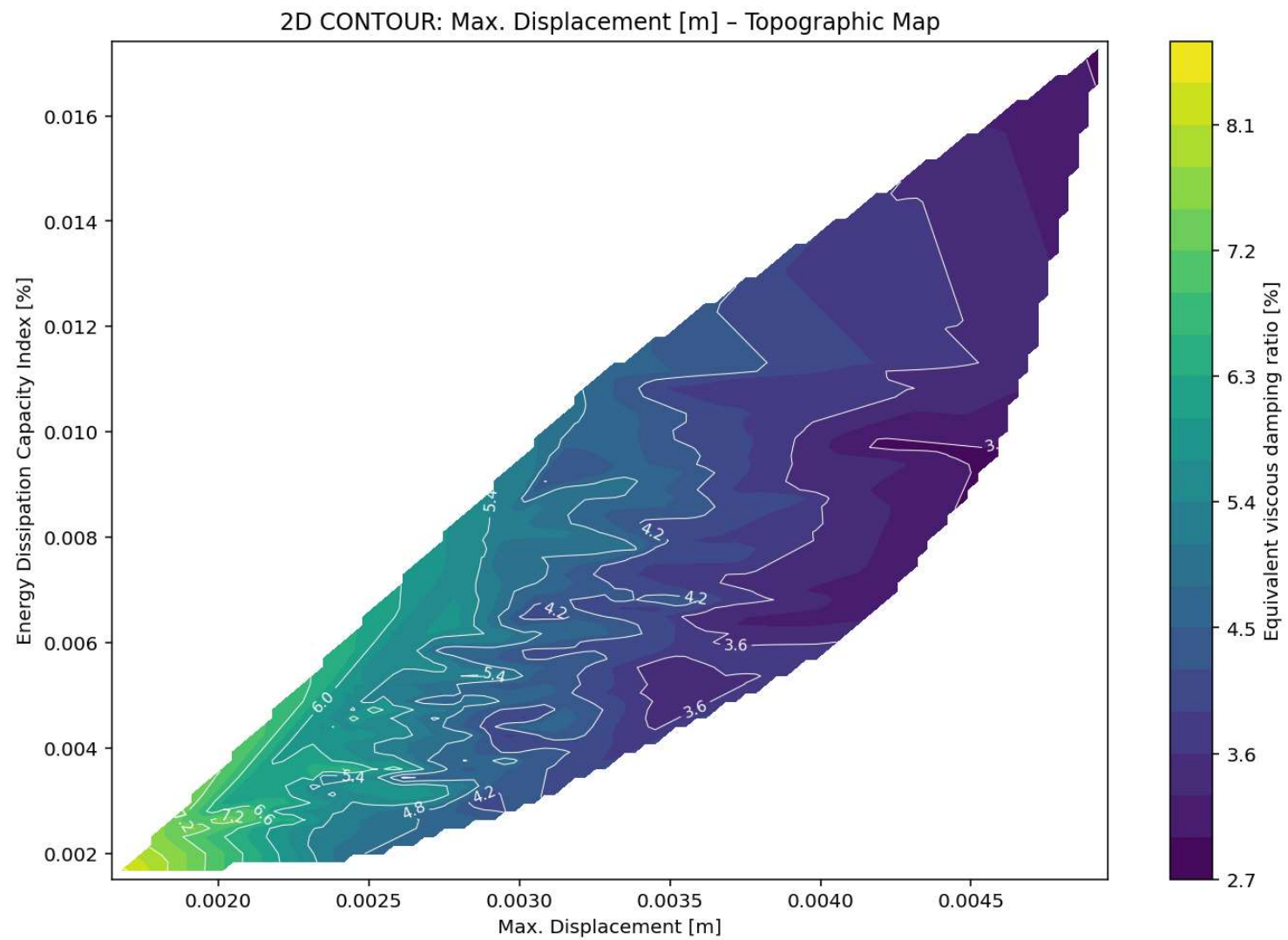
3D CONTOUR: Max. Displacement [m] - Max. Velocity [m/s] - Max. Acceleration [m/s²]



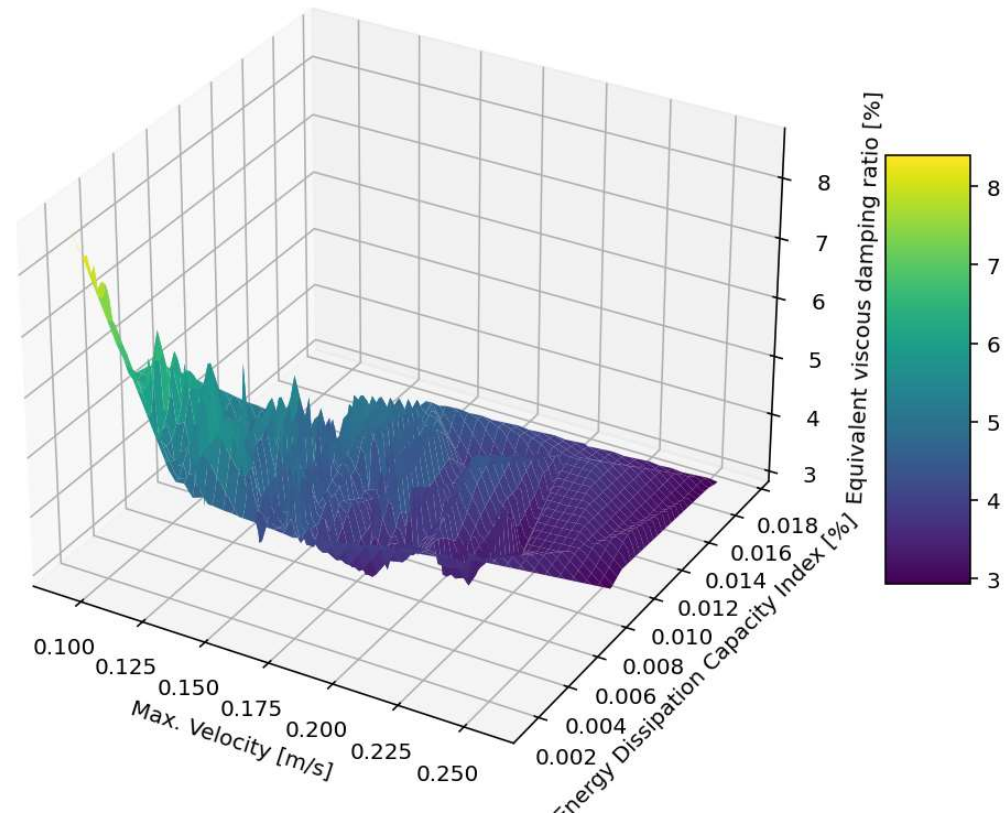


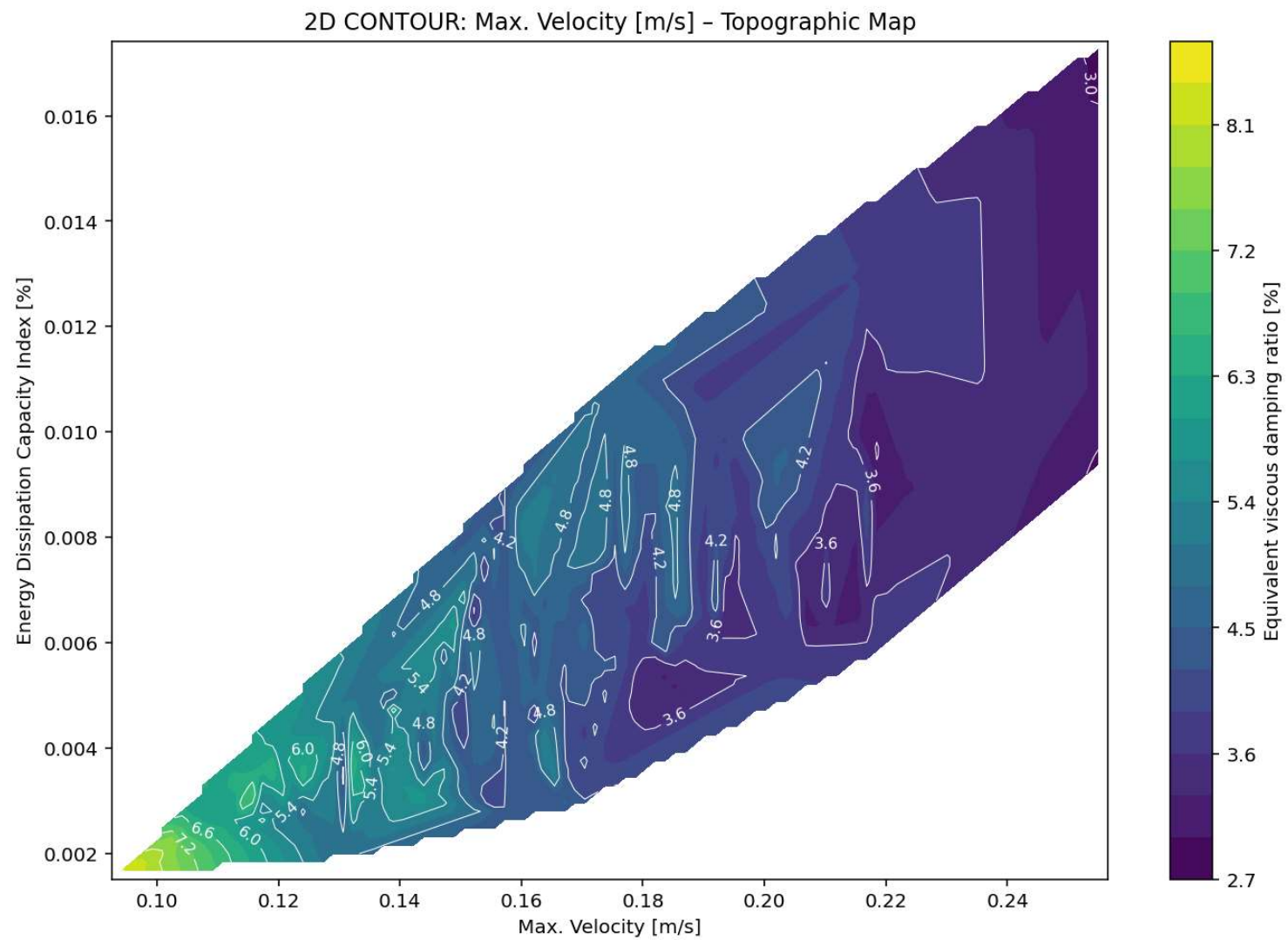
3D CONTOUR: Max. Displacement [m] - Energy Dissipation Capacity Index [%] - Equivalent viscous damping ratio [%]





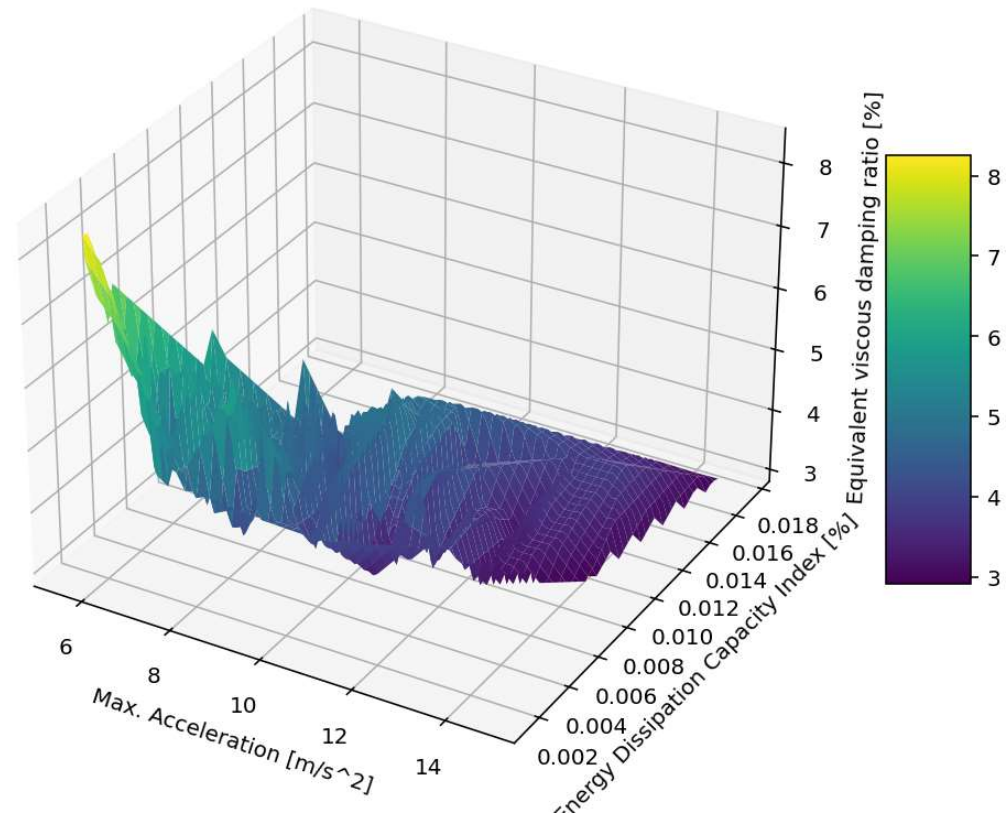
3D CONTOUR: Max. Velocity [m/s] - Energy Dissipation Capacity Index [%] - Equivalent viscous damping ratio [%]



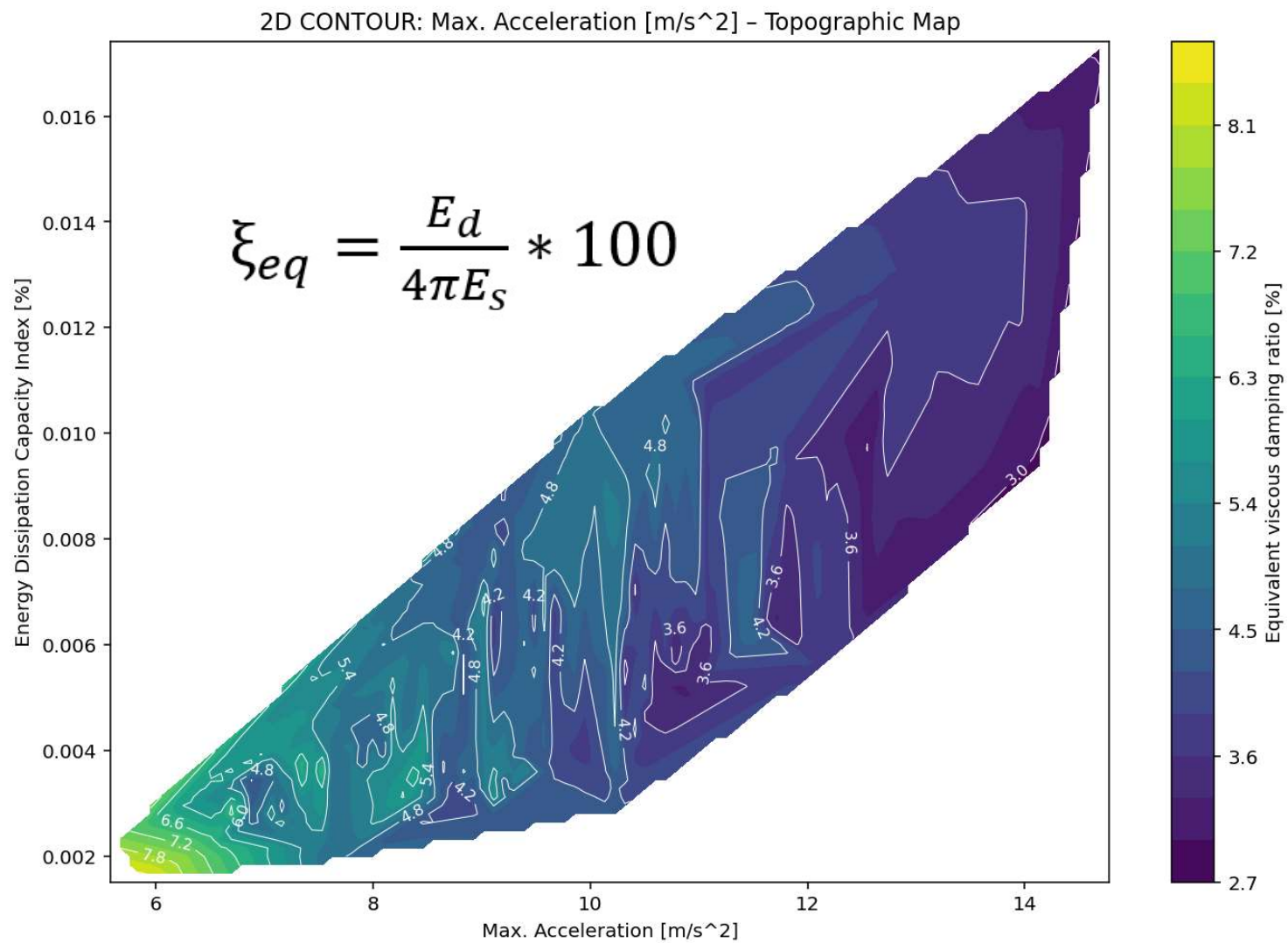


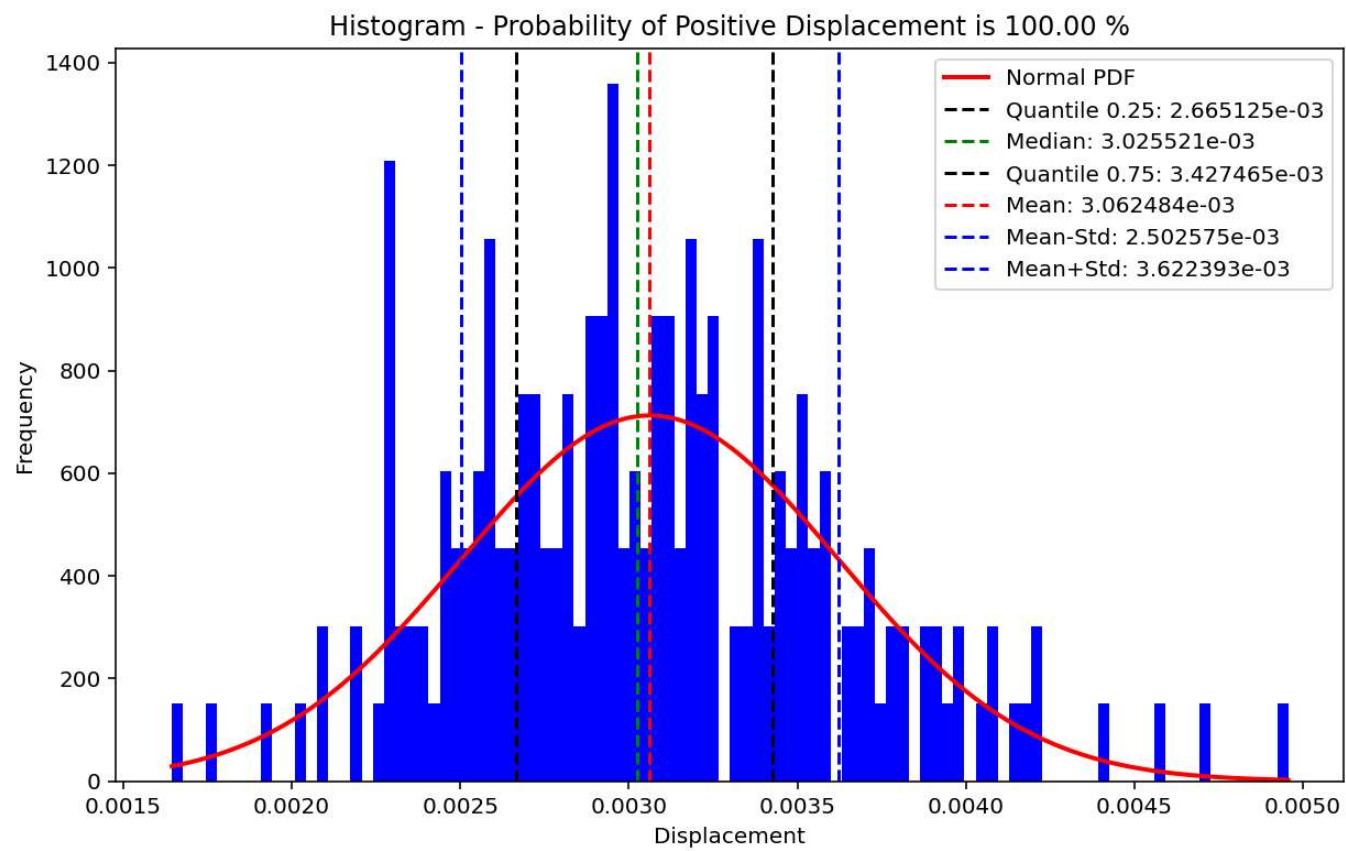
3D CONTOUR: Max. Acceleration [m/s²] - Energy Dissipation Capacity Index [%] - Equivalent viscous damping ratio [%]

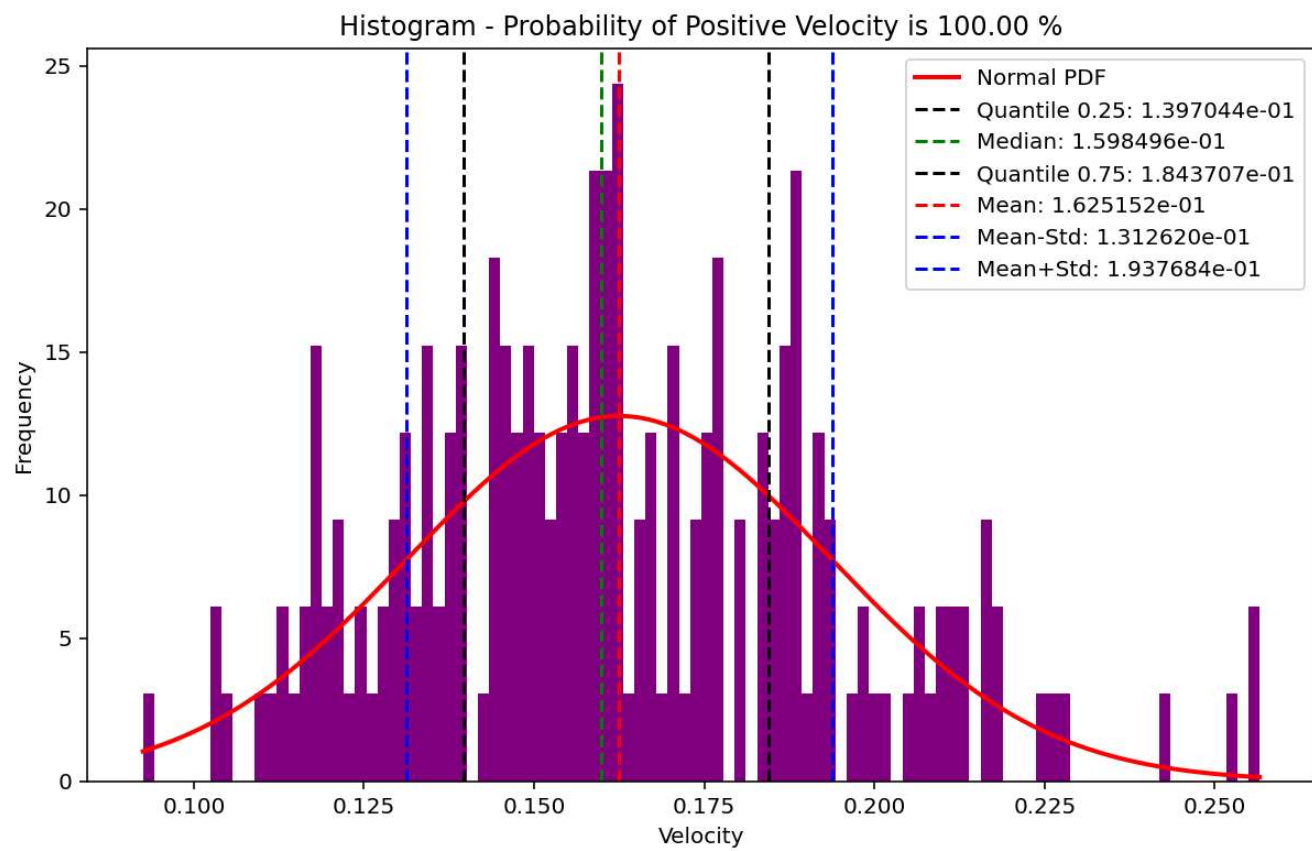
$$\xi_{eq} = \frac{E_d}{4\pi E_s} * 100$$

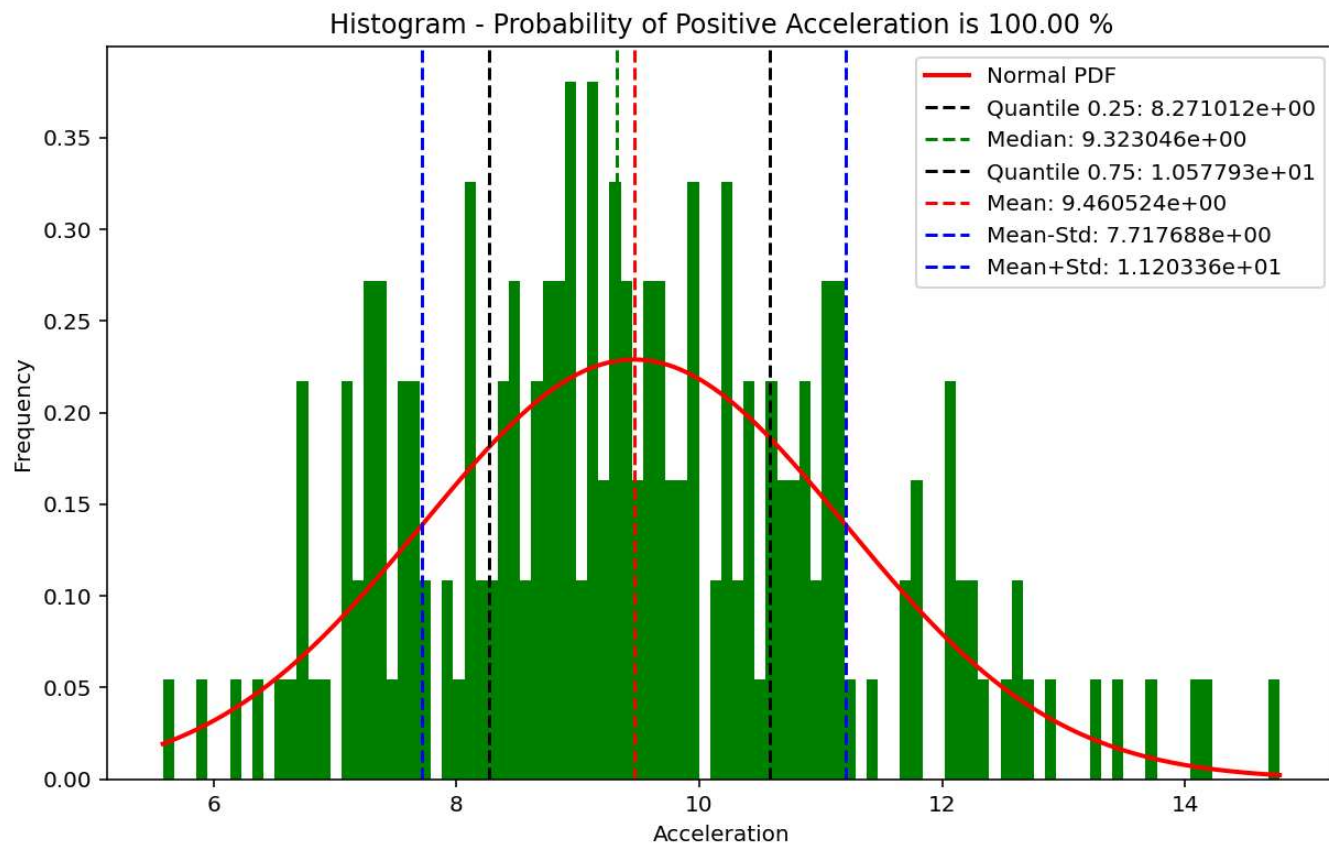


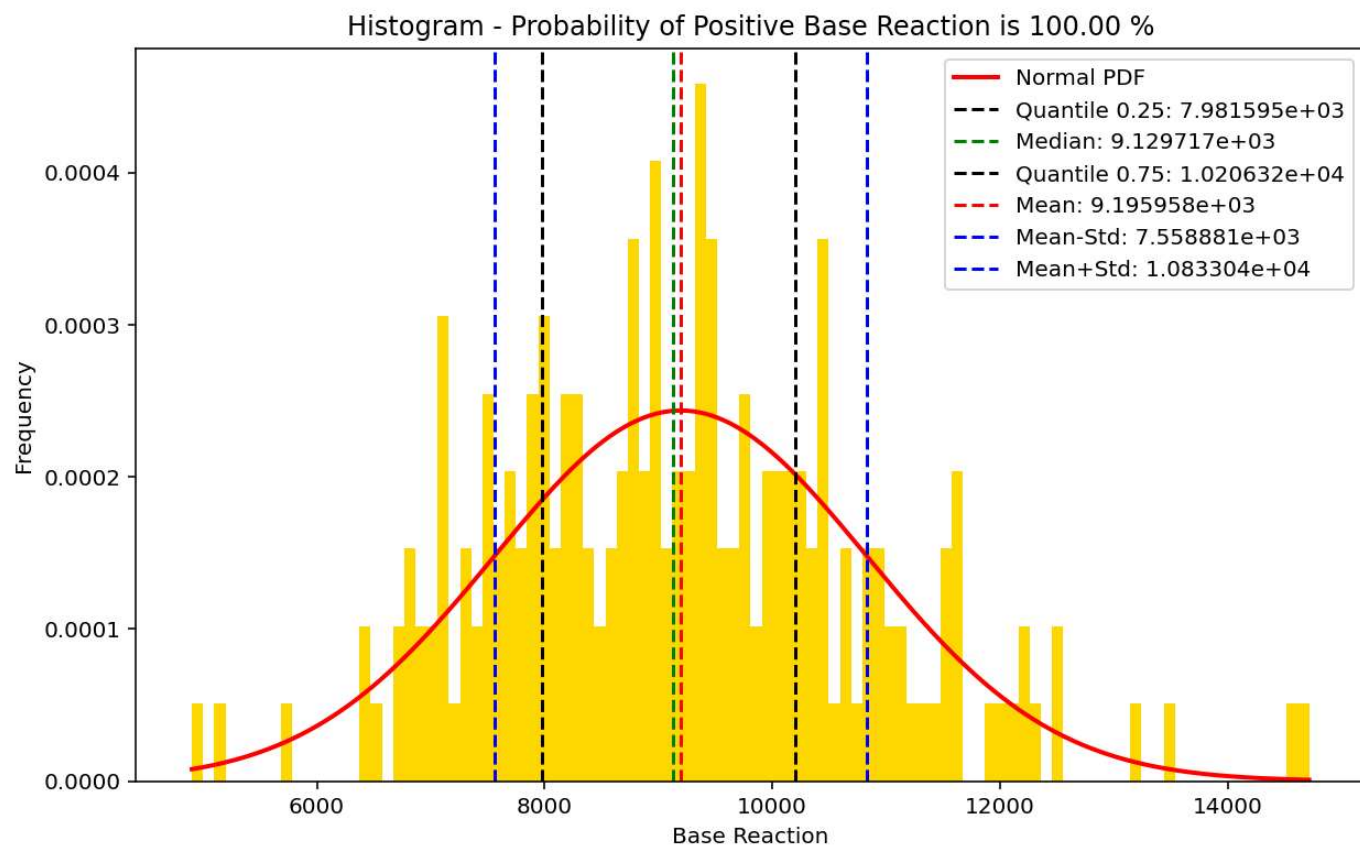
$$\text{Energy Dissipation Capacity Index} = \frac{E_d(\text{Dissipated Energy from Earthquake})}{E_d(\text{Dissipated Energy from Cyclic Displacement})} * 100$$

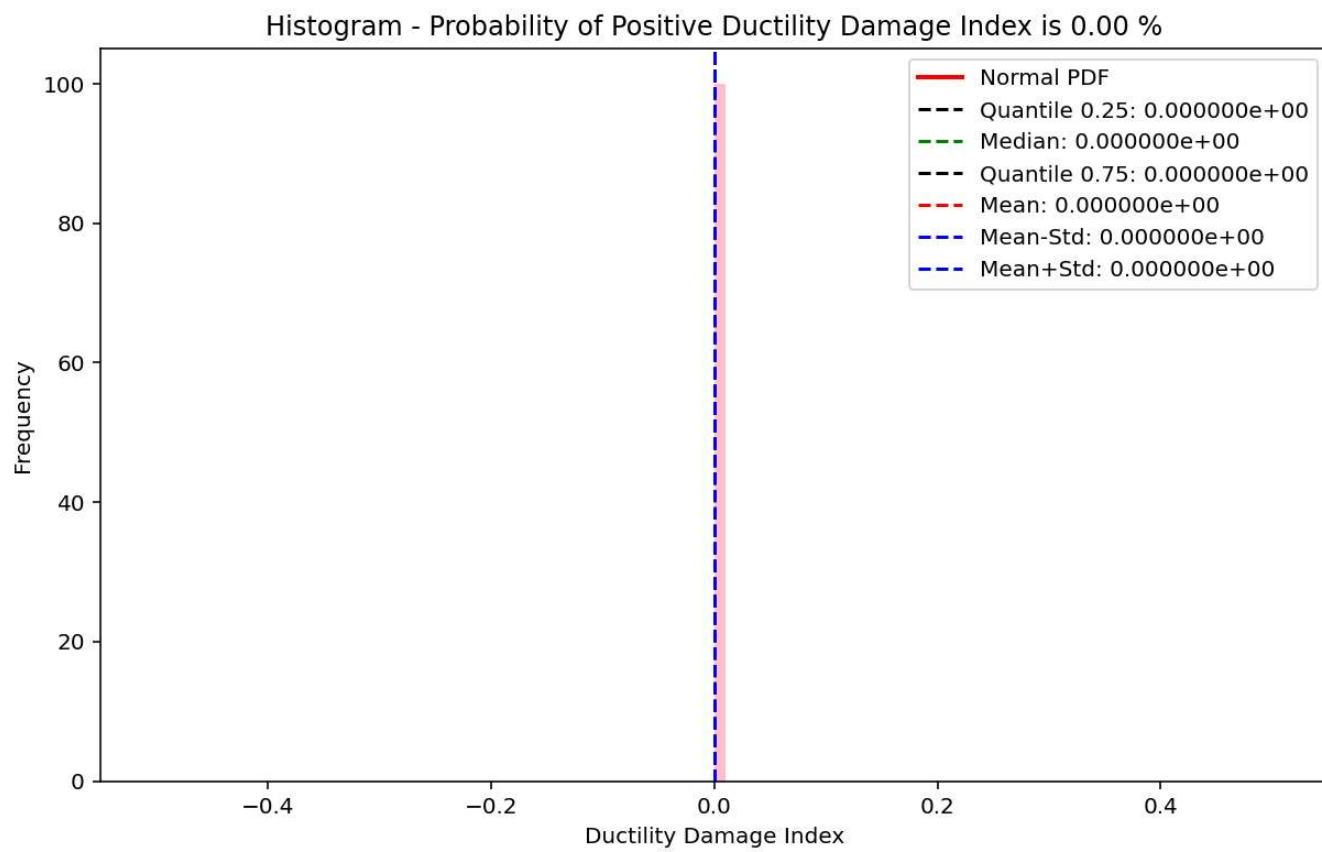


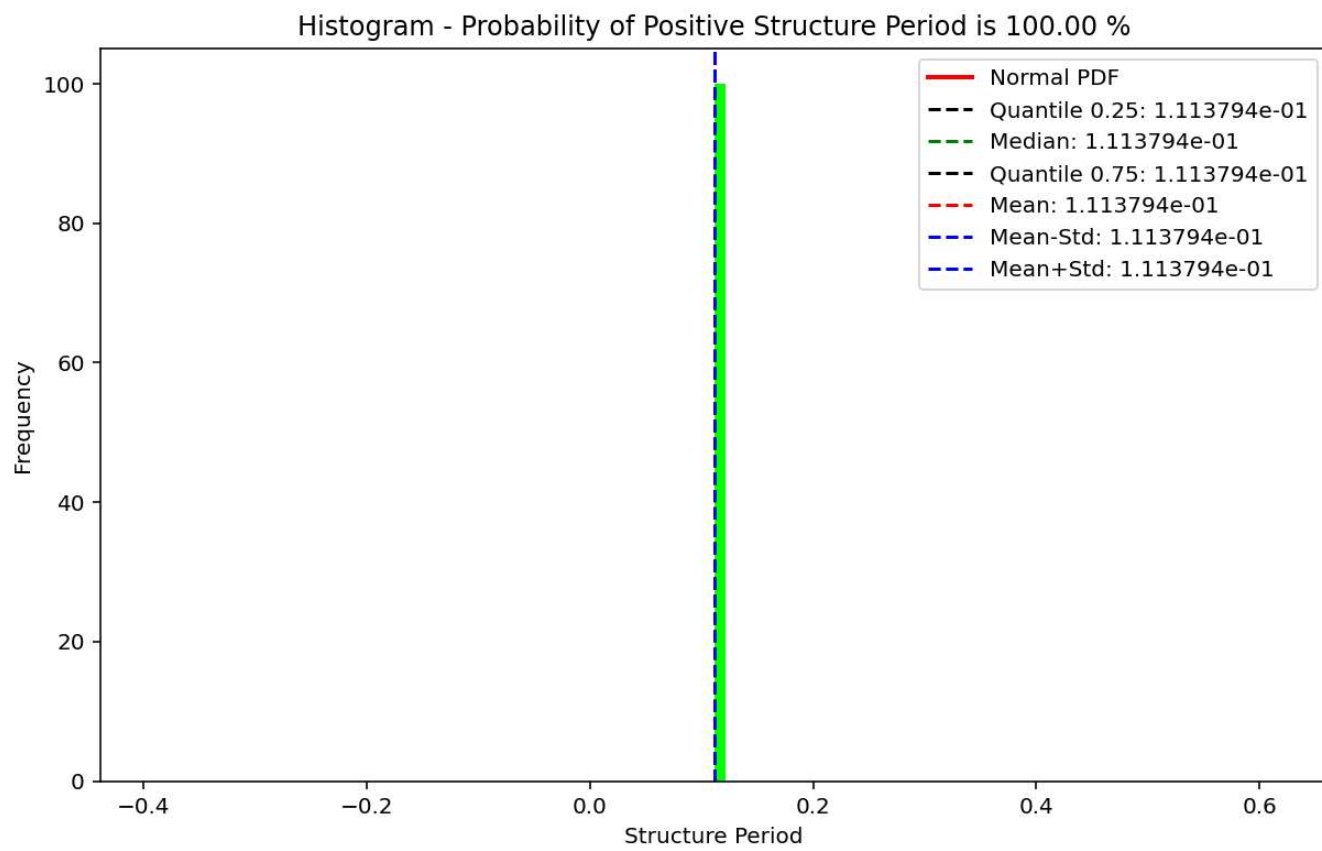


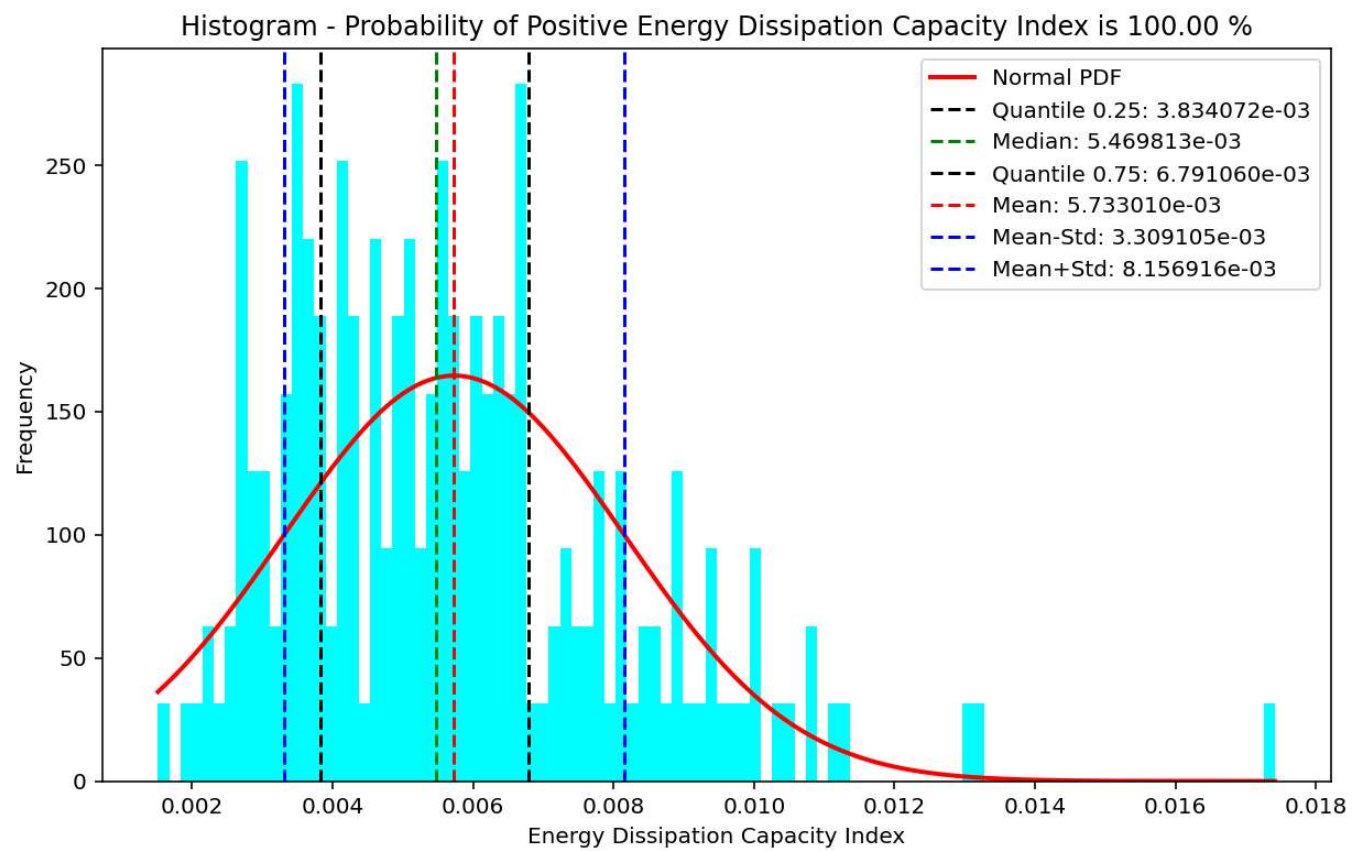


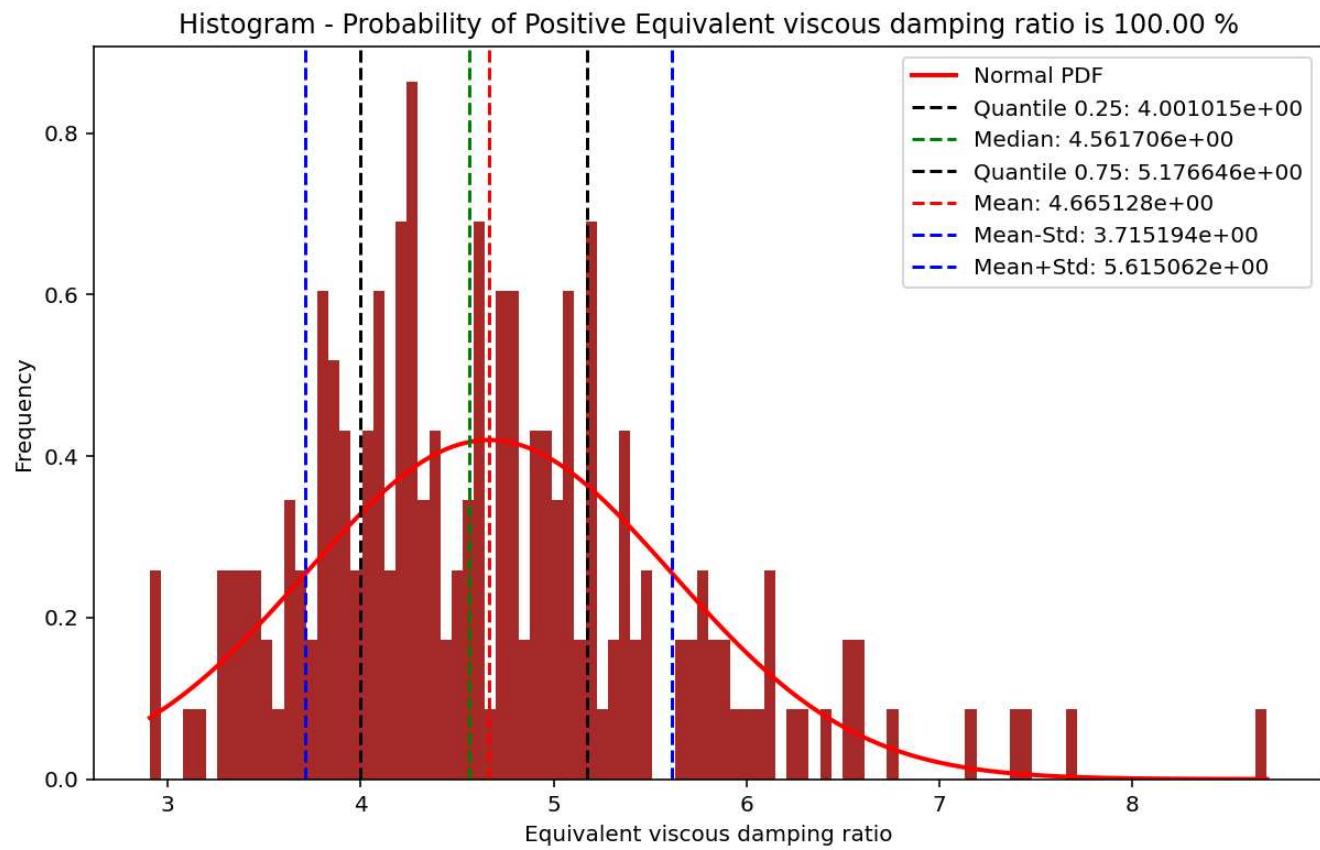


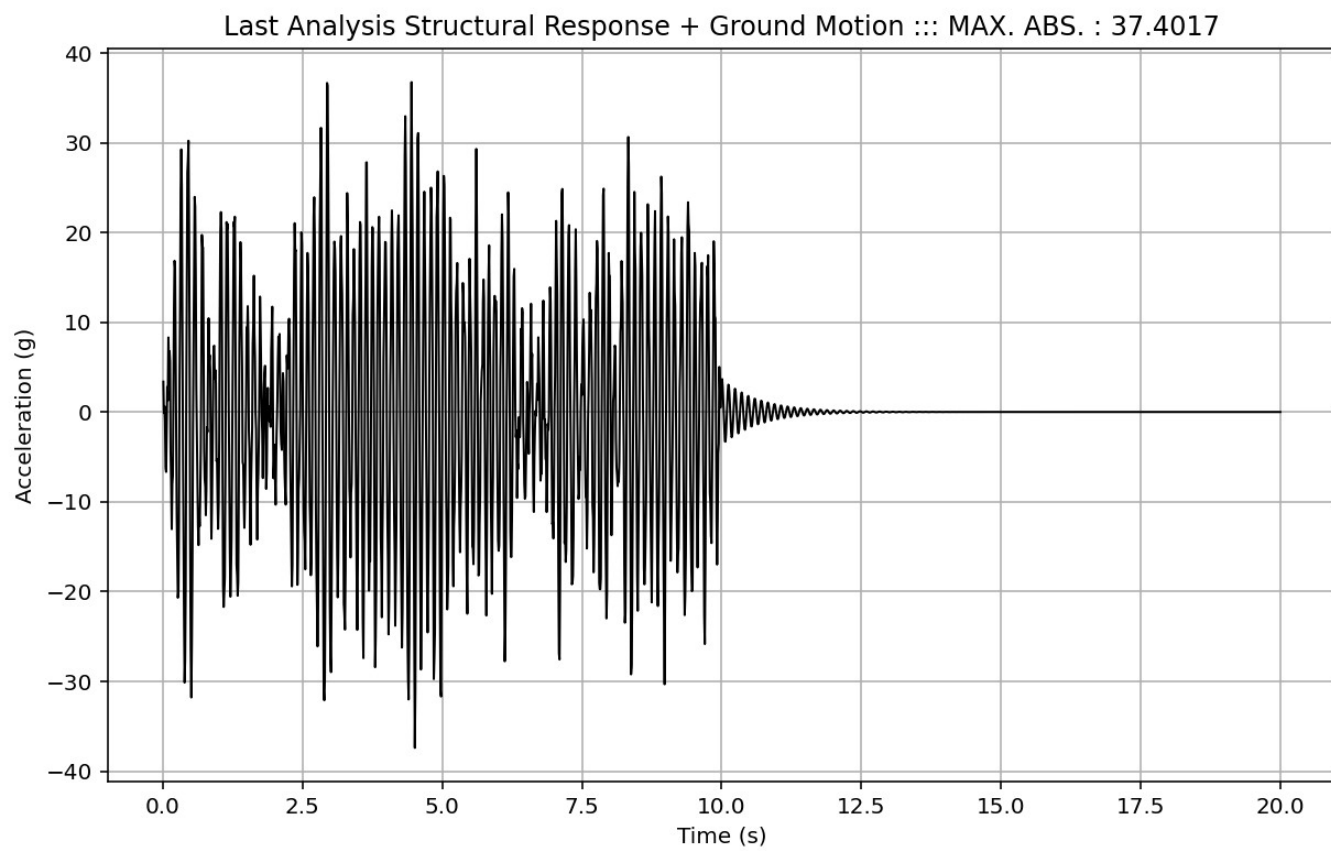


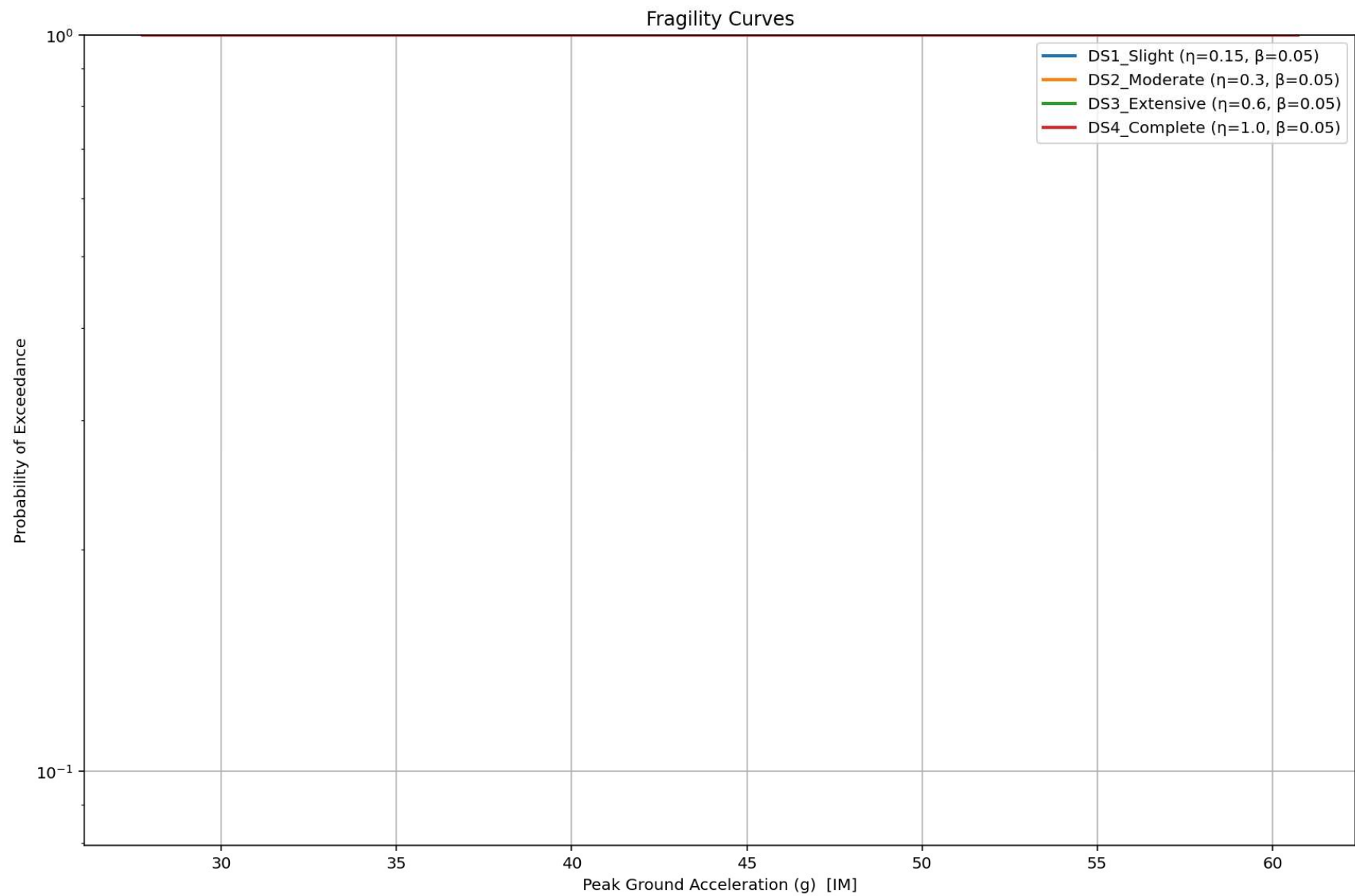




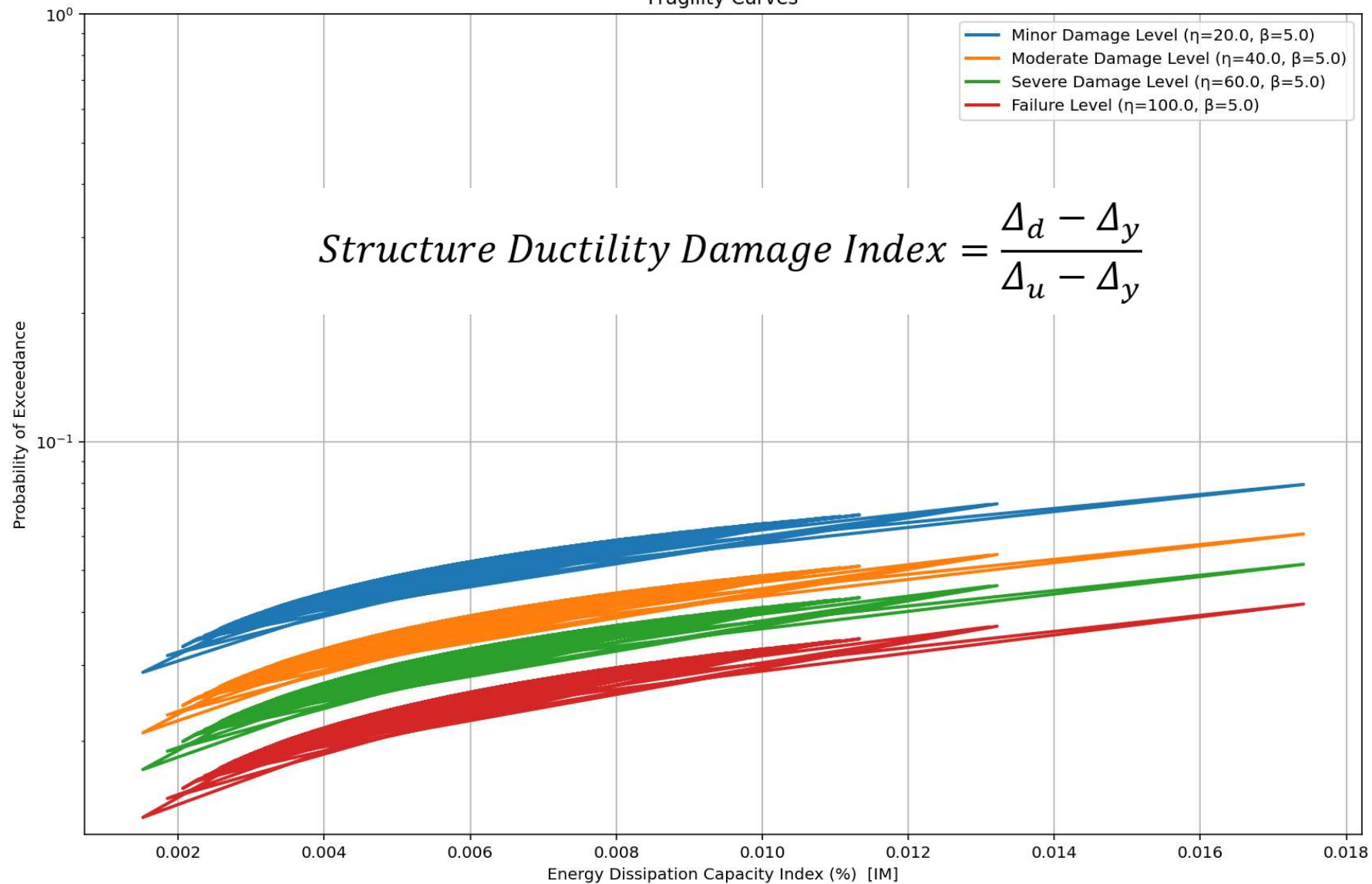


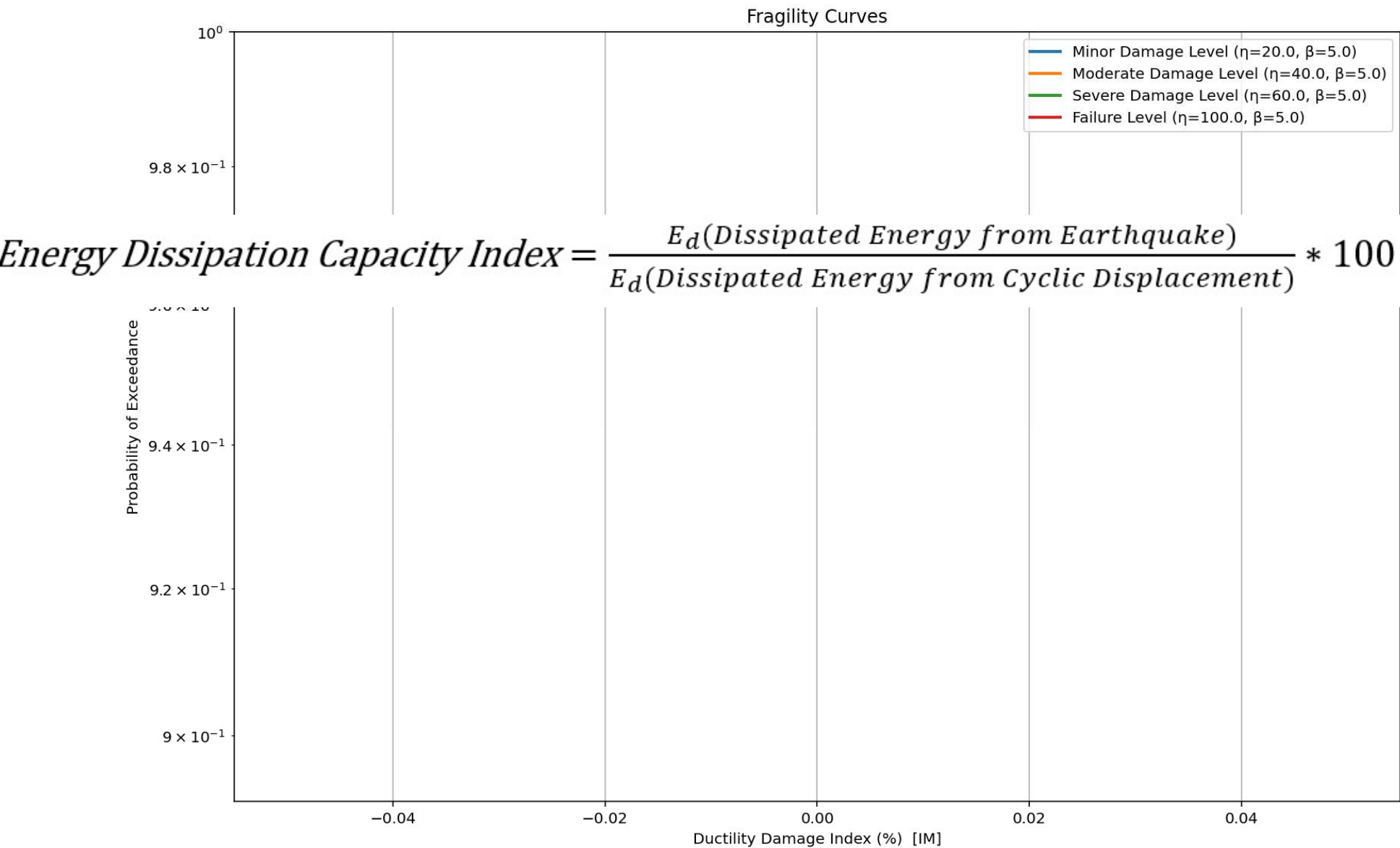




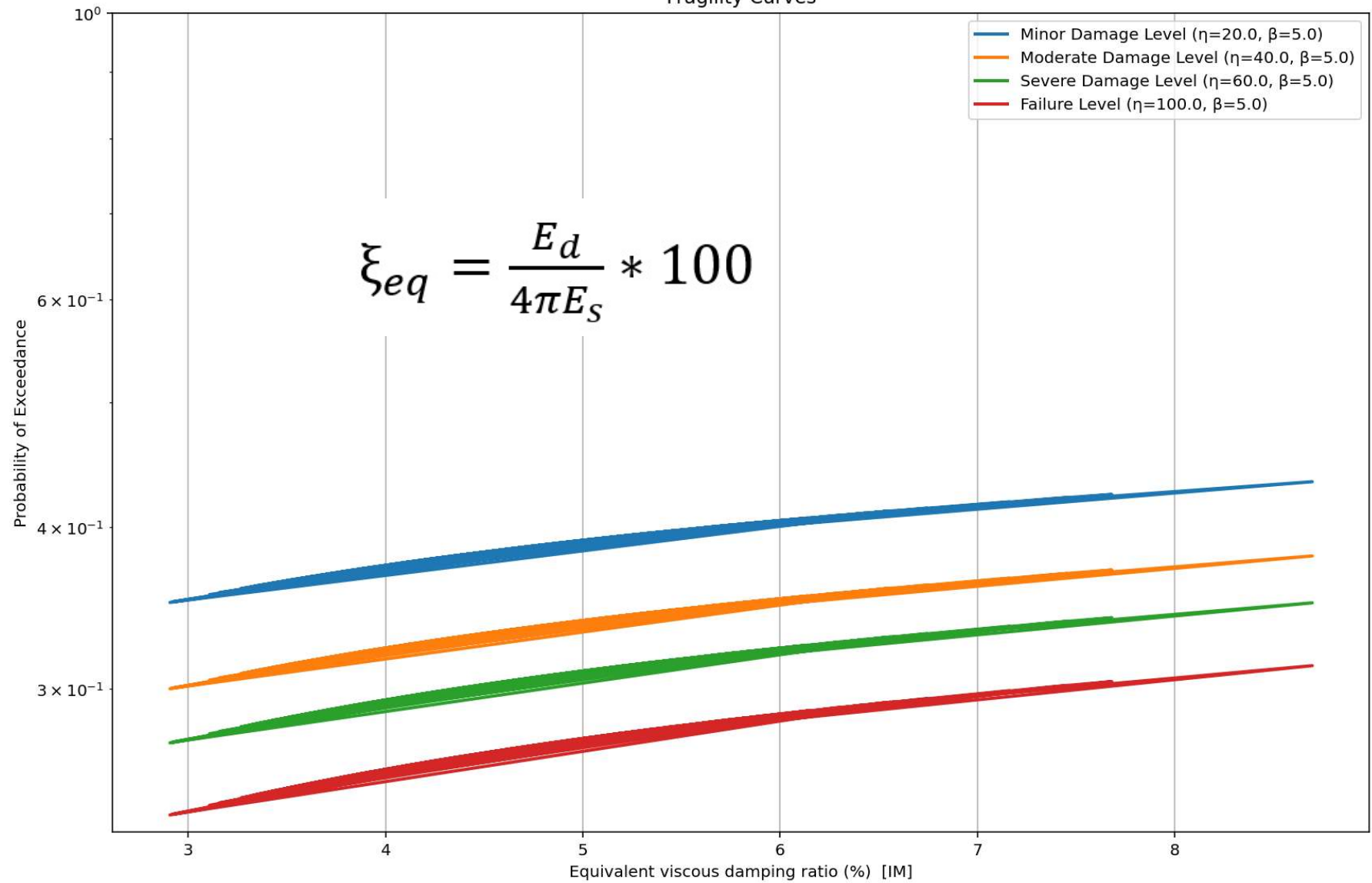


Fragility Curves

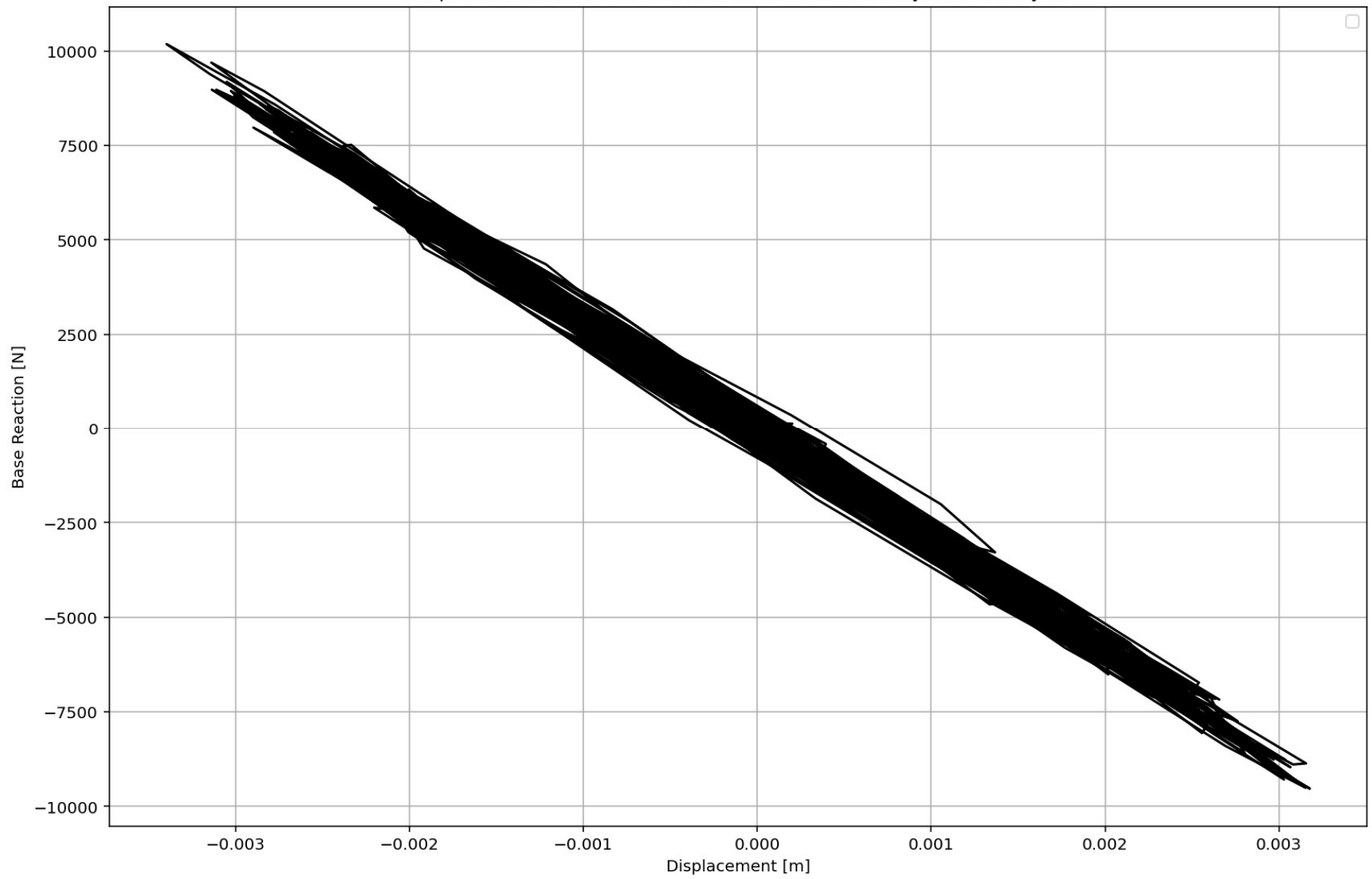


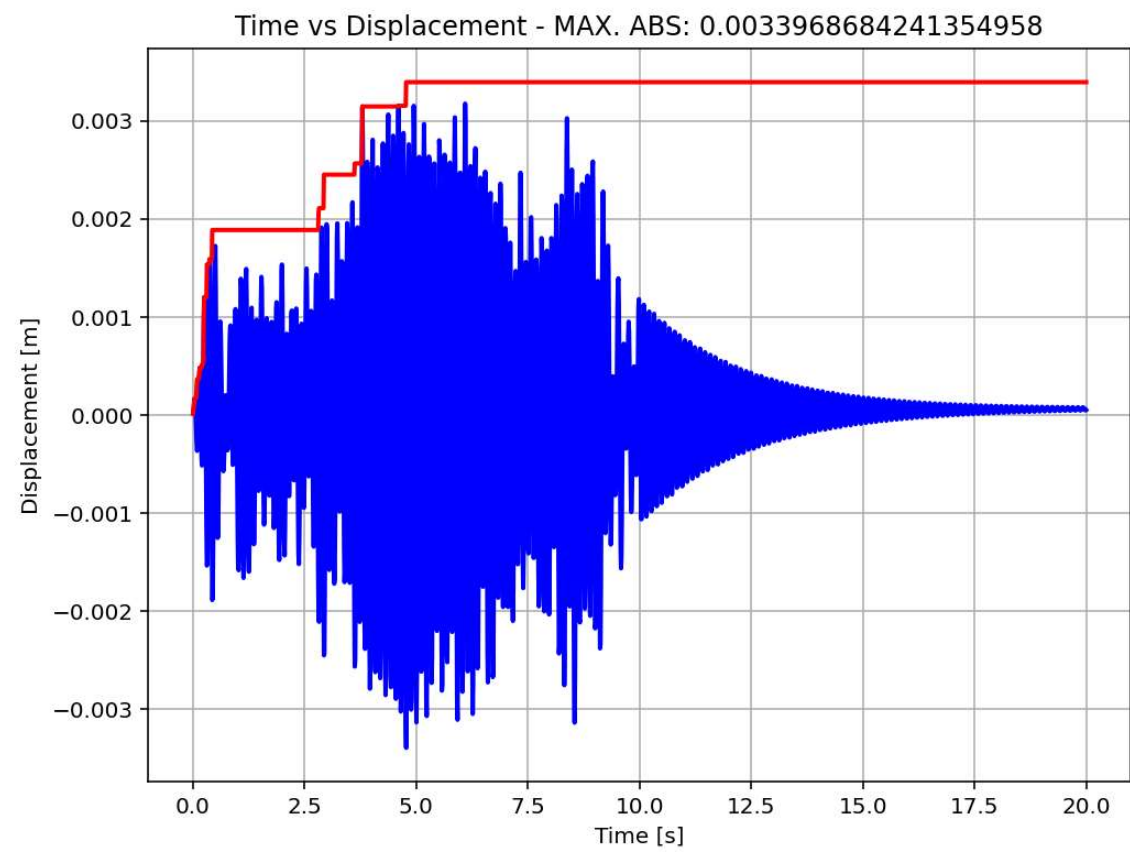


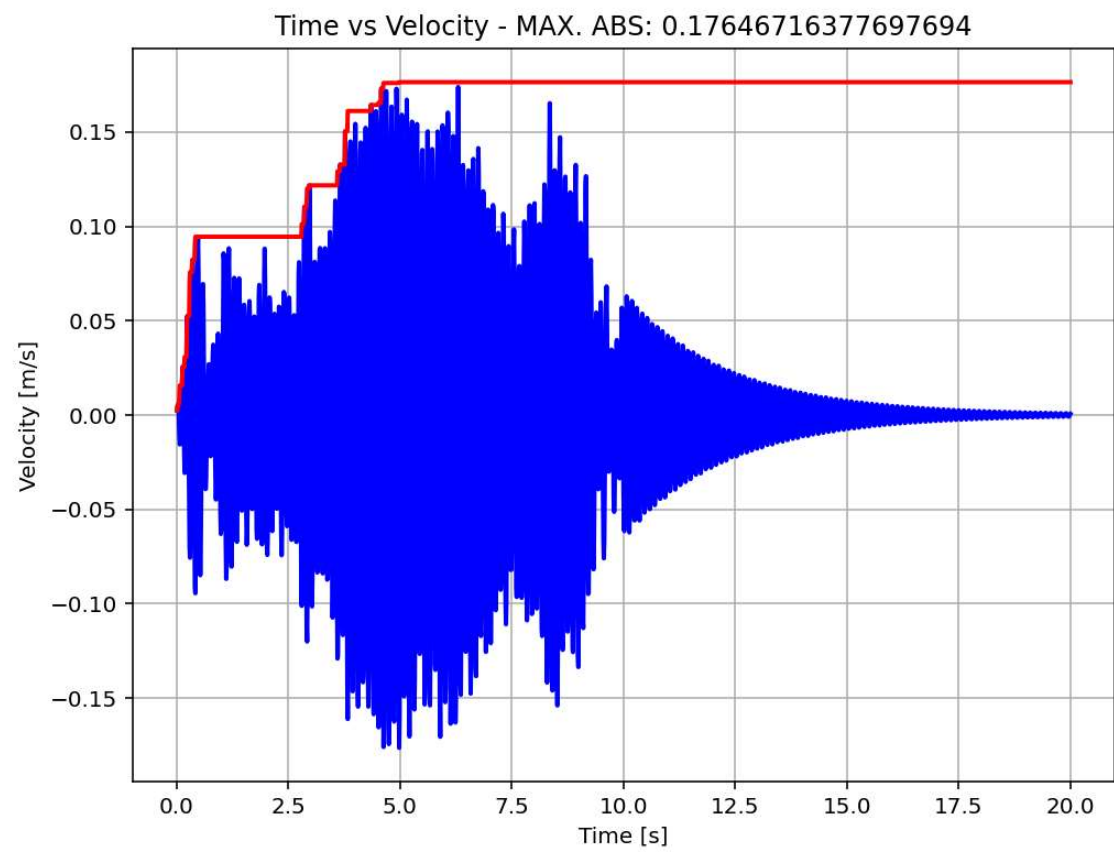
Fragility Curves

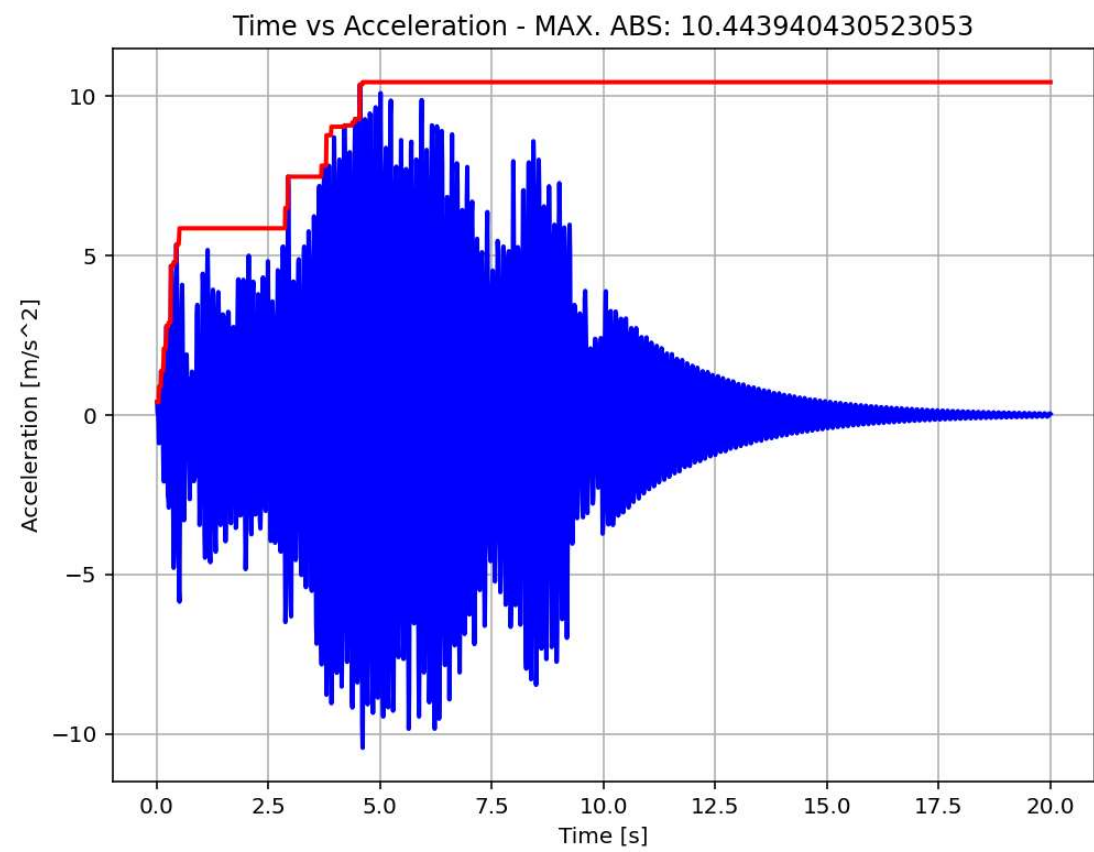


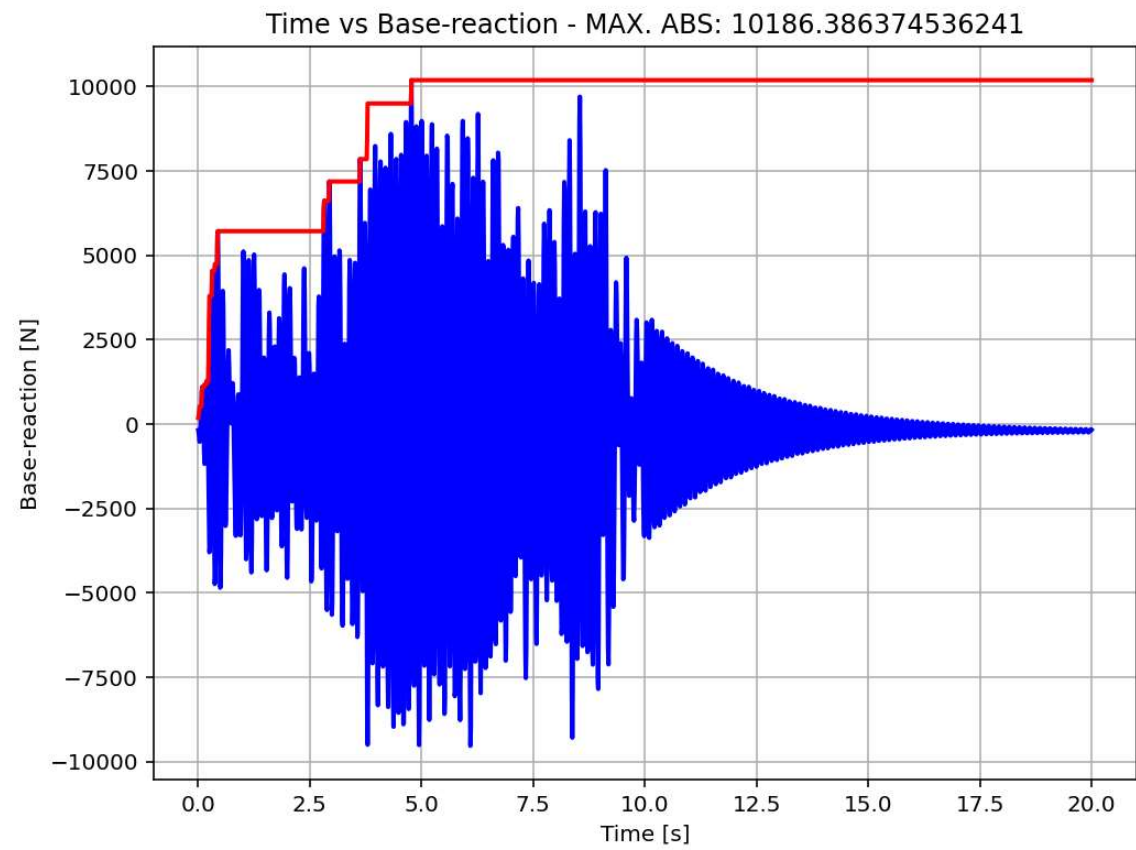
Displacement & Base Reaction Relation From Last Dynamic Analysis

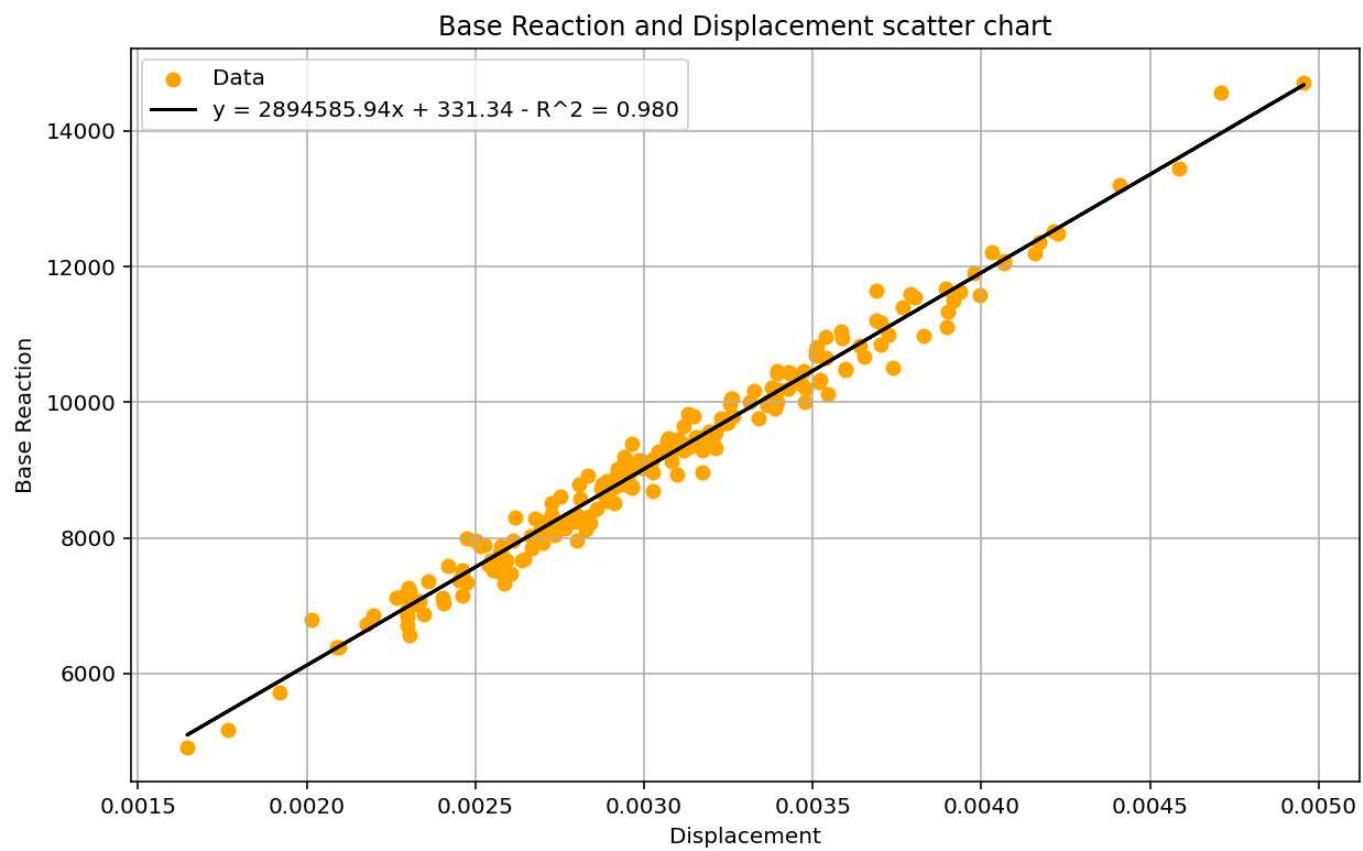


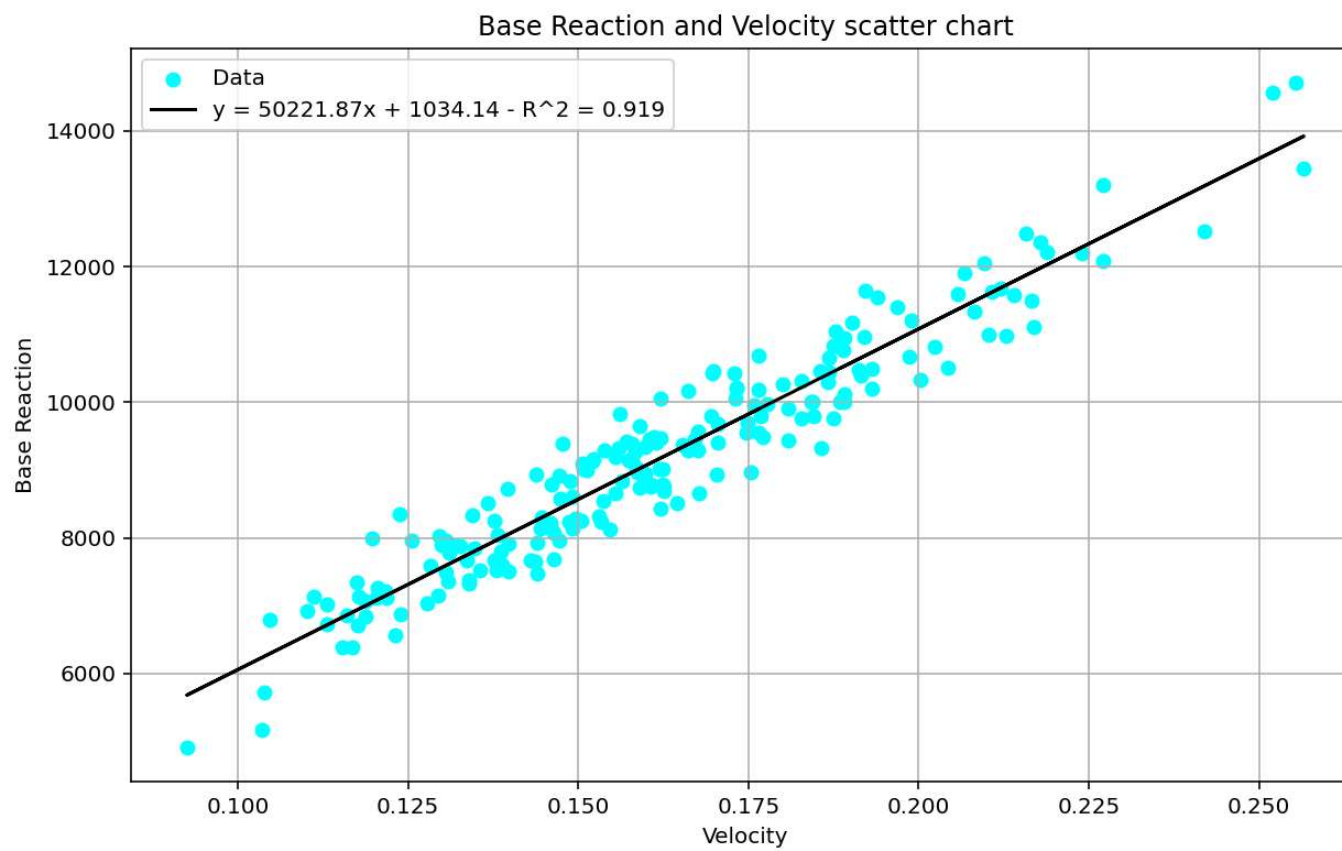


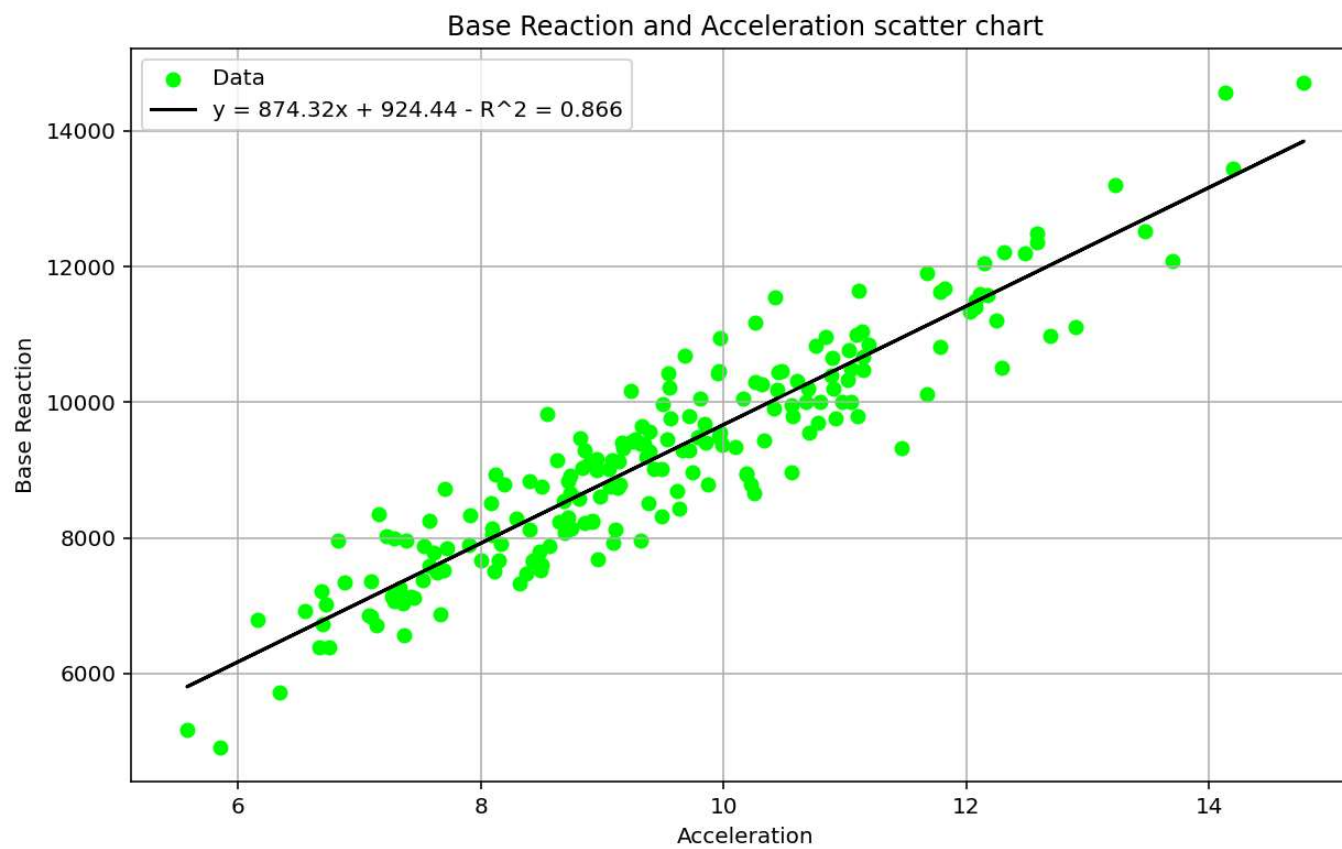


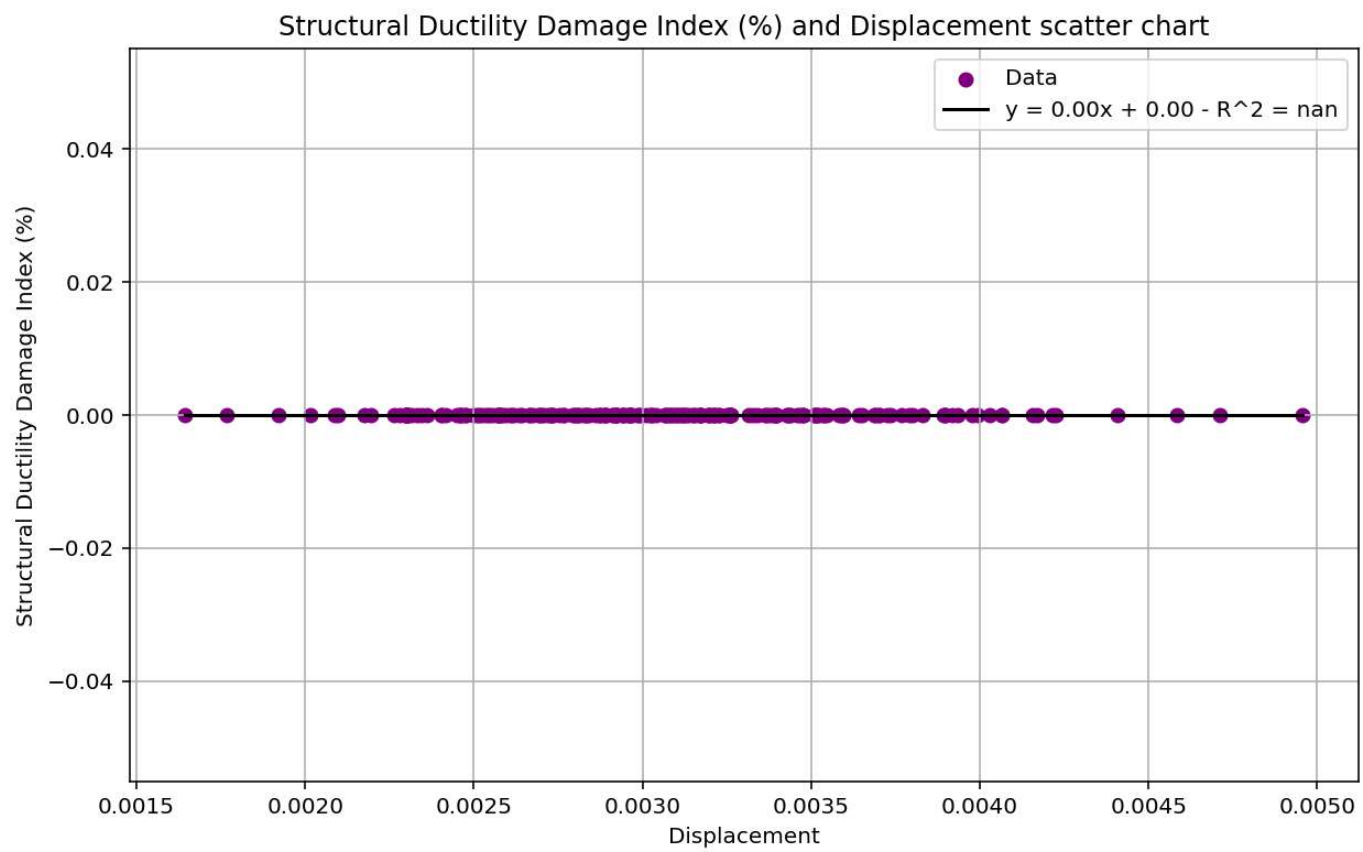


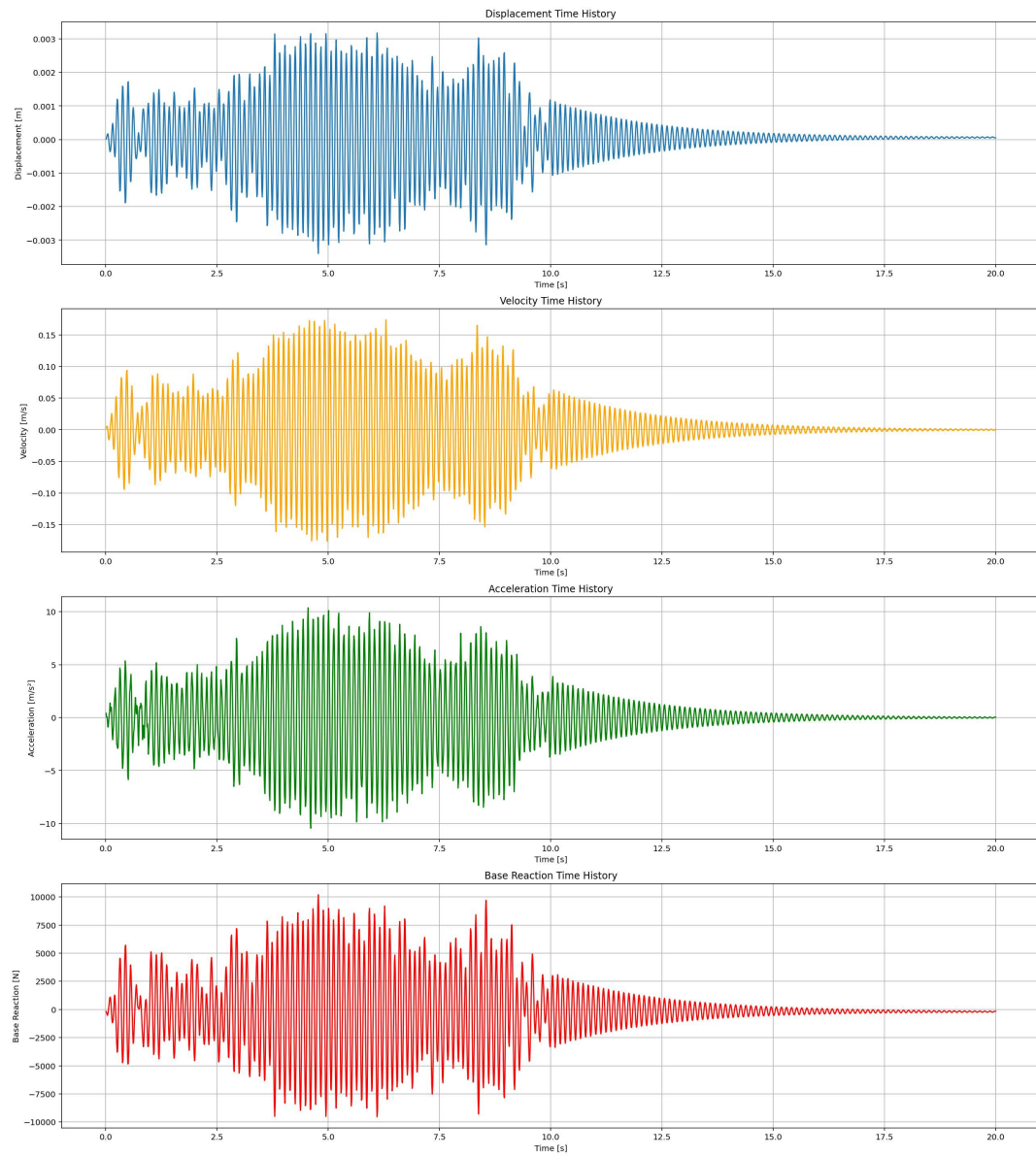


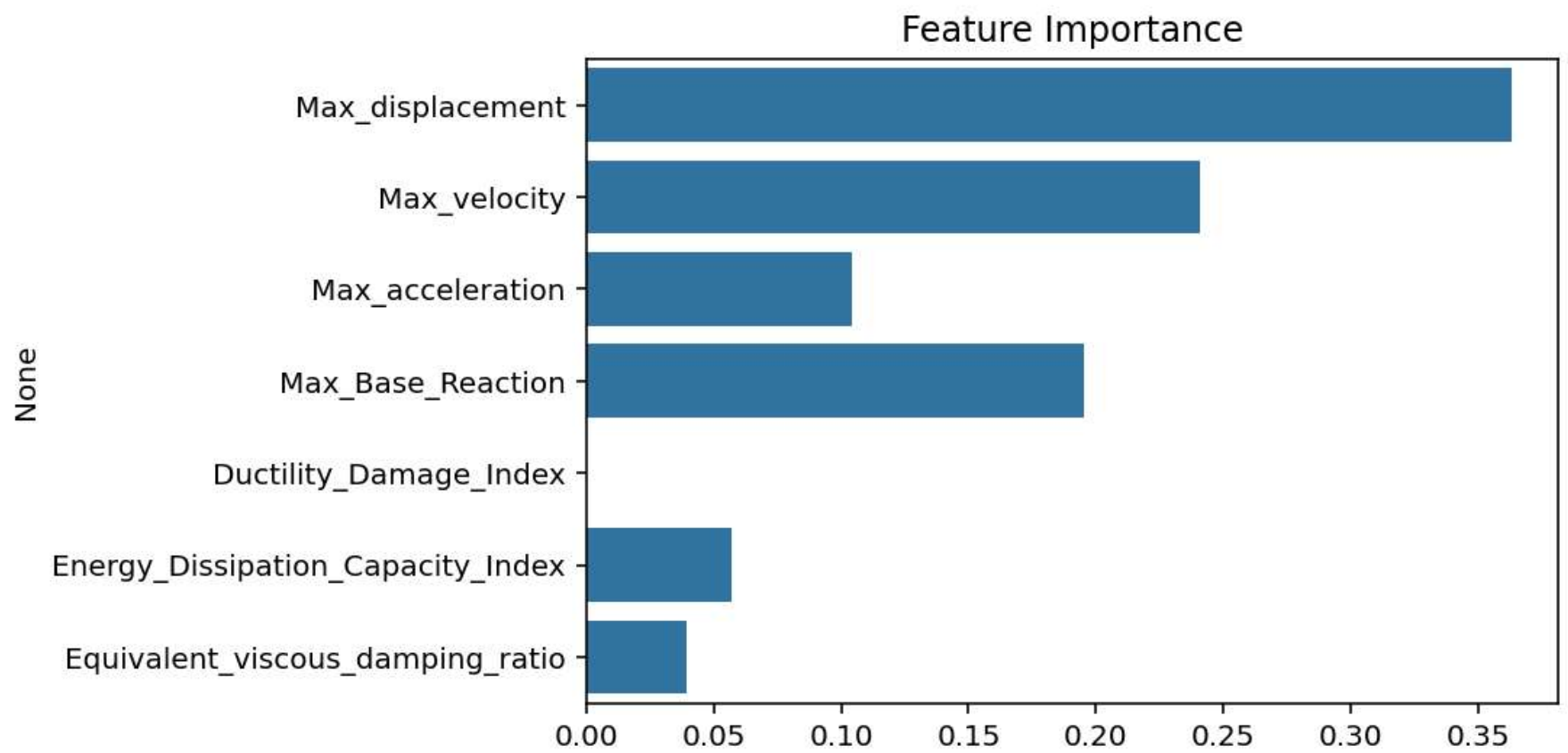












Correlation Heatmap

